

1) Quadratics

a) Solving by factoring

Example 1

Solve a) $6x^2 + 11x - 35 = 0$
 b) $20x^2 + 80 = 82x$.

Always look to see if we can first divide each term by a common factor.

a) $6x^2 + 11x - 35 = 0$
 $(3x - 5)(2x + 7) = 0$
 $x = \frac{5}{3}$ or $x = -\frac{7}{2}$ ← Solve $(3x - 5) = 0$ and $(2x + 7) = 0$.

b) $20x^2 + 80 = 82x$
 $20x^2 - 82x + 80 = 0$ ← Rearrange as $ax^2 + bx + c = 0$.
 $10x^2 - 41x + 40 = 0$ ← Divide through by any integer common factor.
 $(5x - 8)(2x - 5) = 0$
 $x = \frac{8}{5}$ or $x = \frac{5}{2}$

Example 2

A rectangle has length $(x + 3)$ metres and width $(2x + 1)$ metres.
 The area of the rectangle is 12 m^2 .
 Find the length of the rectangle.



Area = $(2x + 1)(x + 3) = 2x^2 + 7x + 3 = 12$
 $2x^2 + 7x - 9 = 0$
 $(2x + 9)(x - 1) = 0$
 $x = -\frac{9}{2}$ or $x = 1$

But the solution cannot be $x = -\frac{9}{2}$ as this will lead to a negative length and width, so $x = 1$.
 Length = $x + 3 = 1 + 3 = 4$ metres.

b) Solving inequalities

- | | | |
|------------------------------|-------------------------------|---------------------------------|
| 2. a) $x^2 + 7x + 12 \leq 0$ | b) $x^2 - x - 30 < 0$ | c) $x^2 + 2x - 48 > 0$ |
| d) $x^2 - 5x + 6 \geq 0$ | e) $2x^2 - 11x + 12 > 0$ | f) $3x^2 + 14x - 5 \leq 0$ |
| g) $15 + 2x - x^2 \leq 0$ | h) $16 - 6x - x^2 < 0$ | i) $21 - x - 2x^2 > 0$ |
| j) $10x^2 + 43x + 28 \geq 0$ | k) $x^2 - 9 \geq 0$ | l) $50 - 2x^2 < 0$ |
| 3. a) $x^2 - 2x > 35$ | b) $x^2 + 6 \leq 5x$ | c) $x^2 \leq x + 20$ |
| d) $x(x + 3) \geq 10$ | e) $x^2 < 9x + 10$ | f) $x^2 - 4(x + 6) > 21$ |
| g) $24 > 11x - x^2$ | h) $7 + 2(4x^2 - 15x) \leq 0$ | i) $\frac{x^2 + 12}{2} \geq 4x$ |
| j) $(x + 5)^2 > 1$ | | |

c) Solving by quadratic equation

Example 13

Solve the inequality $2 - 6x - x^2 \leq 0$. Leave your answer in surd form.

Consider $2 - 6x - x^2 = 0$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(-1)(2)}}{2(-1)}$$

$$= \frac{6 + \sqrt{44}}{-2} \text{ or } \frac{6 - \sqrt{44}}{-2}$$

$$= \frac{6 + 2\sqrt{11}}{-2} \text{ or } \frac{6 - 2\sqrt{11}}{-2}$$

$$= -3 - \sqrt{11} \text{ or } -3 + \sqrt{11}$$

Now $2 - 6x - x^2 \leq 0$

$$x \leq -3 - \sqrt{11} \text{ or } x \geq -3 + \sqrt{11}$$

$$a = -1$$

$$b = -6$$

$$c = 2$$

$$\sqrt{44} = \sqrt{4 \times 11} = 2\sqrt{11}$$

**Example 14**

Solve the equation $2x^2 = 5x - 1$. Write your answer correct to 2 decimal places.

When asked to solve to 2 decimal places, it is likely that the quadratic formula will be appropriate.

$$2x^2 - 5x + 1 = 0$$

$$a = 2, b = -5, c = 1$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(1)}}{2(2)}$$

$$= \frac{5 + \sqrt{17}}{4} \text{ or } \frac{5 - \sqrt{17}}{4}$$

$$= 2.28 \text{ or } 0.22 \text{ (2 d.p.)}$$

We must first rearrange to $ax^2 + bx + c = 0$.

Q In examinations, unless otherwise indicated, answers to all questions on all topics should be rounded to 3 significant figures.

However, do **not** round your answers to 3 significant figures until the final answer.

A significant number of candidates lose marks in examinations by rounding earlier in the question which then leads to an inaccurate answer.

You should therefore keep all numbers unrounded until the final answer.

Exercise 1.4

1. Solve each of these quadratic equations. Write your answers correct to 2 decimal places.

a) $2x^2 - 3x - 4 = 0$

b) $5x^2 - 11x + 4 = 0$

c) $3x^2 + 12x + 5 = 0$

d) $x^2 + 5x - 2 = 0$

e) $4x^2 + 2x - 5 = 0$

f) $x^2 + x - 4 = 0$

g) $3x^2 = 6x - 2$

h) $9x = 6x^2 + 2$

i) $3 - 8x = 2x^2$

j) $x^2 + 3x = 1$

2. Use the formula to solve each of these quadratic equations. Leave your answers in surd form.

a) $2x^2 - x - 5 = 0$

b) $3x^2 - 6x + 1 = 0$

c) $5x^2 - 3x - 7 = 0$

d) $6x^2 - x - 4 = 0$

e) $x^2 + 8x - 3 = 0$

f) $x^2 + 7x + 3 = 0$

g) $x^2 + 10x - 5 = 0$

h) $2x^2 + 5x - 4 = 0$

i) $4x^2 = 5 - 10x$

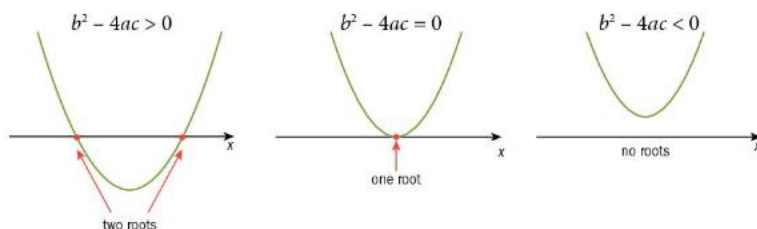
j) $4x - 1 - x^2 = 0$

3. Use the formula to solve the following inequalities. Leave your answers in surd form.

d) Discriminant of quadratic equation

The discriminant is the value of $b^2 - 4ac$ in $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

- If $b^2 - 4ac > 0$ there are two distinct **real roots**.
 If $b^2 - 4ac = 0$ there are equal roots (one repeated root).
 If $b^2 - 4ac < 0$ there are no real roots.



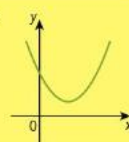
Example 18

Work out whether each of these quadratic equations has two distinct roots, equal roots or no real roots.

- a) $x^2 - 3x + 5 = 0$
 b) $3x^2 + x - 6 = 0$
 c) $25x^2 + 20x + 4 = 0$

Note: We know from 1.3 that if $y = ax^2 + bx + c$ and $a > 0$, then the curve is $\cup$ shaped.

If $b^2 - 4ac < 0$, then there are no real roots and the curve does not cross the x -axis, so y is positive for all values of x .



- | | |
|--|--------------------|
| a) $b^2 - 4ac = (-3)^2 - 4(1)(5) = 9 - 20 = -11 < 0$ | no real roots |
| b) $b^2 - 4ac = (1)^2 - 4(3)(-6) = 1 + 72 = 73 > 0$ | two distinct roots |
| c) $b^2 - 4ac = (20)^2 - 4(25)(4) = 400 - 400 = 0$ | equal roots |

- Show that $1 + 100x^2 - 20x \geq 0$ for all values of x .
- If $2x^2 - ax + 8 = 0$ has no real roots, find the range of possible values of a .
- If $6 - 2x - kx^2 = 0$ has a repeated root, find the value of k .
- The equation $x^2 + px + q = 0$, where p and q are constants, has roots -3 and 2 . Find the value of p and q .
- Use the discriminant to find the nature of the roots of the equation $3x + 4 = \frac{5}{x}$.
- The quadratic equation $kx^2 + 5x + 2 = 0$ has two distinct real roots. Find the range of possible values of k .
- Find the value of p for which the quadratic equation $px^2 - 4px + 2 - p = 0$ has equal roots.
- Prove that the quadratic equation $(q - 5)x^2 + 5x - q = 0$ has real roots for any value of q .
- The quadratic equation $x + k + \frac{9}{x} = 0$ has equal roots. Find the two possible values of k .
- The equation $px^2 + qx + r = 0$, where p , q and r are constants, has roots $-\frac{1}{2}$ and $\frac{3}{4}$. Find the smallest possible integer values of p , q and r .

e) Graphs of quadratic functions

Example 23

Express $y = x^2 - 6x + 10$ in the form $y = (x + p)^2 + q$.

Sketch the curve, stating the coordinates of the vertex.

$$y = (x - 3)^2 - 9 + 10 = (x - 3)^2 + 1$$

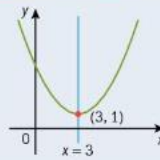
Coordinates of the vertex are (3, 1).

The **minimum** value of y is 1.

$a > 0$ therefore the graph is $\cup$ shaped.

The line of symmetry: $x = \frac{-b}{2a} = \frac{-(-6)}{2(1)}$

$$x = 3$$



$$\begin{aligned} b^2 - 4ac &= (-6)^2 - 4(1)(10) \\ &= -4 < 0 \\ &\text{therefore no real roots.} \end{aligned}$$

Example 24

Express $y = 3 - 8x - 2x^2$ in the form $y = a(x + b)^2 + c$.

Sketch the curve, stating the coordinates of the vertex.

$$\begin{aligned} y &= -2[x^2 + 4x] + 3 \\ &= -2[(x + 2)^2 - 4] + 3 \\ &= -2(x + 2)^2 + 8 + 3 \\ &= -2(x + 2)^2 + 11 \end{aligned}$$

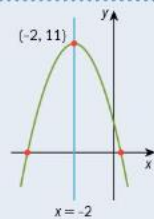
Coordinates of the vertex are (-2, 11).

The **maximum** value of y is 11.

$a < 0$ therefore the graph is $\cap$ shaped.

The line of symmetry: $x = \frac{-b}{2a} = \frac{-(-8)}{2(-2)}$

$$x = -2$$



$$\begin{aligned} b^2 - 4ac &= (-8)^2 - 4(-2)(3) \\ &= 88 > 0 \\ &\text{therefore two distinct roots.} \end{aligned}$$

Example 25

A sheep pen is in the shape of a rectangle.

One of the sides of the pen is a wall.

A farmer puts fencing on the other three sides of the pen.

The two sides that touch the wall are each x metres.

He uses 40 m of fencing.

Find the maximum area of the pen.



The third side of the pen = $40 - 2x$

$$\begin{aligned} \text{Area of the pen} &= x(40 - 2x) \\ &= 40x - 2x^2 \\ &= -2[x^2 - 20x] \\ &= -2[(x - 10)^2 - 100] \\ &= -2(x - 10)^2 + 200 \end{aligned}$$

The **maximum** area is 200 m² (when $x = 10$).

Summary:

To solve quadratic equations

- Factorise and solve.
- Or use the **method of completing the square**.
- Or use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

To solve quadratic inequalities

- Solve the corresponding equation $ax^2 + bx + c = 0$ to find the values of x .
- Sketch the graph of $y = ax^2 + bx + c$.
- Write down the solutions using inequality signs.

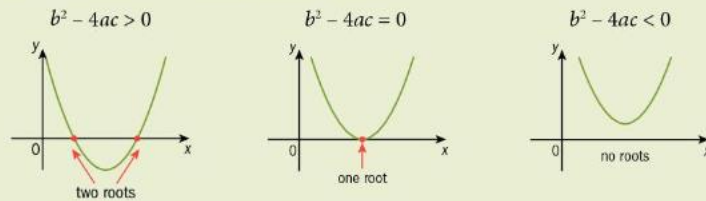
The discriminant of the quadratic equation $ax^2 + bx + c = 0$

- Discriminant $= b^2 - 4ac$

If $b^2 - 4ac > 0$ there are two distinct real roots.

If $b^2 - 4ac = 0$ there are equal roots (one repeated root).

If $b^2 - 4ac < 0$ there are no real roots.



$ax^2 + bx + c = 0$, where a , b and c are constants, $a \neq 0$.

Graphs of quadratic equations $y = ax^2 + bx + c$ where a , b and c are constants, $a \neq 0$

- If $a > 0$, the curve is $\cup$ shaped and has a **minimum** value.
- If $a < 0$, the curve is $\cap$ shaped and has a **maximum** value.
- The **equation of the line of symmetry** is $x = \frac{-b}{2a}$.

To solve simultaneous equations where one is linear and one is quadratic

- Rearrange the linear equation to write x in terms of y or y in terms of x .
- Then substitute this into the quadratic equation.

Recognise and solve equations in x which are quadratic in some function of x

- e.g. To solve $x^6 - 2x^3 + 1 = 0$, let $y = x^3$ and solve $y^2 - 2y + 1 = 0$.

2) Functions and transformations

a) Composite functions

Example 6

The functions f , g , and h are defined by

$$f: x \mapsto 4x - 1, x \in \mathbb{R} \quad g: x \mapsto \frac{1}{x+2}, x \neq -2 \quad h: x \mapsto (2-x)^2, x \in \mathbb{R}$$

Find **a)** $fg(x)$ **b)** $hh(x)$

$$\begin{aligned} \text{a) } fg(x) &= f\left(\frac{1}{x+2}\right) \\ &= 4\left(\frac{1}{x+2}\right) - 1 \\ &= \frac{4 - 1(x+2)}{x+2} \\ &= \frac{2-x}{x+2}, x \neq -2 \end{aligned}$$

Substitute $g(x)$ into $f(g(x))$.

$f: x \mapsto 4x - 1$

Simplify using a common denominator of $x + 2$.

$$\begin{aligned} \text{b) } hh(x) &= h[(2-x)^2] \\ &= [2 - (2-x)^2]^2 \\ &= [2 - (4 - 4x + x^2)]^2 \\ &= (-2 + 4x - x^2)^2 \end{aligned}$$

$h: x \mapsto (2-x)^2$ subtract from 2 and then square.

Example 7

The functions f and g are defined by

$$f: x \mapsto ax^2, x \in \mathbb{R} \quad g: x \mapsto 3x + b, x \in \mathbb{R} \quad \text{where } a \text{ and } b \text{ are constants.}$$

Given that $gf(1) = 5$ and $gg(2) = 14$ find

a) the values of a and b **b)** the value of $fg(-3)$.

$$\text{a) } gf(1) = g[a(1)^2] = g(a)$$

Substitute $x = 1$ in to $f(x)$.

$$= 3a + b$$

$g: x \mapsto 3x + b$ multiply by 3 and then add b .

$$gf(1) = 5$$

$$\text{So, } 3a + b = 5$$

$$gg(2) = g(6 + b)$$

Substitute $x = 2$ in to $g(x)$.

$$= 3(6 + b) + b$$

$$gg(2) = 14$$

$$\text{So, } 18 + 4b = 14$$

Solve for b .

$$b = -1$$

$$3a - 1 = 5$$

Substitute for $b = -1$ in $3a + b = 5$.

$$a = 2$$

$$\text{b) } f(x) = 2x^2 \quad g(x) = 3x - 1$$

$$fg(-3) = f(3 \times -3 - 1) = f(-10)$$

First work out $g(-3)$.

$$f(-10) = 2(-10)^2 = 200$$

Then substitute this value for x in to f .

b) Inverse functions

Example 9

The function f is defined by $f: x \mapsto x^2 - 6x + 2$ for $0 \leq x \leq 3$.

- Express $f(x)$ in the form $(x + a)^2 + b$, where a and b are constants.
- State the range of f .
- State the domain of $f^{-1}(x)$.
- Find an expression for $f^{-1}(x)$.
- Express $ff^{-1}(x)$ in terms of x .
- Sketch on the same diagram the graphs of $y = f(x)$ and $y = f^{-1}(x)$, making clear the relationship between the graphs.

a) $x^2 - 6x + 2 = (x - 3)^2 - 9 + 2$
 $= (x - 3)^2 - 7$

Use the method of completing the square.

b) Range of f : $-7 \leq f(x) \leq 2$

When $x = 0$, $f(x) = 2$; when $x = 3$, $f(x) = -7$ (which is the minimum value from (a) so $f(x)$ cannot be less than this).

c) Domain of $f^{-1}(x)$: $-7 \leq x \leq 2$

The domain of f^{-1} is always the same as the range of f .

d) Let $y = (x - 3)^2 - 7$

Let $f(x) = y$.

$$x = (y - 3)^2 - 7$$

Change x to y and y to x .

Then, $x + 7 = (y - 3)^2$

Hence, $y = 3 \pm \sqrt{x + 7}$

Rearrange to get y in terms of x .

$f^{-1}: x \mapsto 3 + \sqrt{x + 7}$

However, f^{-1} must be one-to-one, so we must find whether a negative or positive square root is appropriate.

Since $f(0) = 2$, we know that $f^{-1}(2) = 0$.

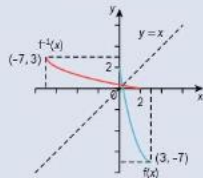
This means that we need the negative square root giving

$f^{-1}: x \mapsto 3 - \sqrt{x + 7}, -7 \leq x \leq 2$

e) $ff^{-1}(x) = f(3 - \sqrt{x + 7})$
 $= (3 - \sqrt{x + 7} - 3)^2 - 7$
 $= (-\sqrt{x + 7})^2 - 7$
 $= x + 7 - 7$
 $= x$

$ff^{-1}(x)$ is always the same as x .

f)



$f^{-1}(x)$ is the reflection of $f(x)$ in the line $y = x$.

11. a) The function $f: x \mapsto 2x^2 - 12x + 8$ is defined for $x \in \mathbb{R}$.
- Express $f(x)$ in the form $a(x + b)^2 + c$, where a , b and c are constants.
 - Find the range of $f(x)$.
- b) The function $g: x \mapsto 2x^2 - 12x + 8$ is defined for $x \geq D$.
- Find the smallest value of D for which g has an inverse.
 - For this value of D , find an expression for $g^{-1}(x)$ in terms of x .
12. The function f is defined by $f: x \mapsto 4x - x^2$ for $x \geq 2$.
- Express $4x - x^2$ in the form $a - (x - b)^2$, where a and b are positive constants.
 - Express $f^{-1}(x)$ in terms of x .
13. A function f is such that $f(x) = 2 + \sqrt{\frac{x-1}{3}}$, for $x \geq 1$.
- Find $f^{-1}(x)$ in the form $ax^2 + bx + c$, where a , b and c are constants.
 - Find the domain of f^{-1} .
14. The functions f and g are defined for $x \in \mathbb{R}$ by
- $$f: x \mapsto 4x - a \qquad g: x \mapsto b + ax$$
- where a and b are constants.
- Given that $f^{-1}(1) = 2$ and $g^{-1}(-1) = -3$, find the values of a and b .

c) Transformations

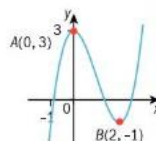
i) Translations

2. a) Sketch the graph of $y = g(x)$, where $g(x) = 2^x$.
- b) On separate diagrams, sketch the graphs of
- $y = g(x + 2)$
 - $y = g(x) - 1$
 - $y = g(x) + 4$.
- Mark on each sketch, where possible, the coordinates of the points where the curve cuts the axes and state the equations of any asymptotes.
3. The graph of $y = f(x)$ is transformed to the graph of $y = f(x + 5)$.
- Describe** the transformation.
4. $f(x) = x(x + 3)$
- Find and simplify the equation of the curve $3 + f(x - 1)$.
 - Describe the transformation.
5. The curve $y = x^2 - x + 1$ is translated by $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$.
- Find and simplify the equation of the translated curve.

ii) Reflections

2. The graph of $y = f(x)$ is transformed to the graph of $y = f(-x) + 7$.
- Describe** the transformation.
3. The diagram shows a sketch of the curve $f(x)$, which passes through the points $A(0, 3)$ and $B(2, -1)$.
- Sketch the graphs of
- $y = -f(x)$
 - $y = f(-x)$
 - $y = -f(x + 4)$
 - $y = 3 - f(-x)$.
- In each case, mark the new position of the points A and B , writing down their coordinates.
4. The points $P(-3, 5)$ and $Q(-2, -8)$ lie on the curve with equation $y = f(x)$.
- Find the coordinates of P and Q after the curve has been transformed by the following transformations:
- $f(-x)$
 - $-f(x)$
 - $f(-x + 1)$
 - $-f(x) - 5$

For example, to sketch the graph of $-f(9 + x)$, first apply the translation of $f(9 + x)$ and then the reflection. To sketch the graph of $f(4 - x)$, rewrite as $f(-(x - 4))$, reflect in the y -axis then translate 4 to the right.



iii) Stretches

2. The diagram shows a sketch of the curve $f(x)$, which passes through the origin, O , and the points $A(-2, 8)$ and $B(1, -1)$.

On separate diagrams, sketch the graphs of

a) $y = 4f(x)$ b) $y = f(-\frac{1}{2}x)$ c) $y = -2f(x + 1)$ d) $y = 1 + f(-2x)$.

Mark the new position of the points O , A and B on each transformation, stating their coordinates.

3. The diagram shows a sketch of the curve $f(x)$.

The curve has a horizontal asymptote with equation $y = 1$ and a vertical asymptote with equation $x = 0$.

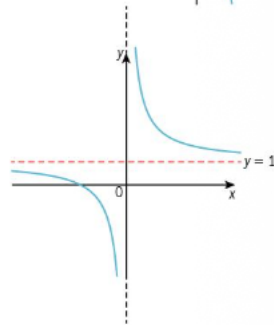
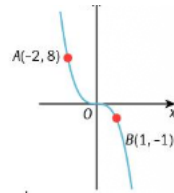
On separate diagrams, sketch the graphs of

a) $y = f(2x)$ b) $y = -3f(x)$ c) $y = 2f(x - 3)$
d) $y = 4 - f(\frac{1}{2}x)$ e) $y = 3 + 4f(x)$ f) $y = 4f(-\frac{1}{2}x)$.

Mark on each sketch the equations of the asymptotes and, if possible, any points where the curve cuts the axes.

4. The graph of $y = f(x)$ is transformed to the graph of $y = -f(2x)$.

Describe fully the two single transformations that have been combined to give the resulting transformation.



Summary:

Chapter summary

Domain and range

- If a function maps x onto $f(x)$, then the values of x is the **domain** of the function and the corresponding values of $f(x)$ is the **range** of the function.

Functions

- A **function** is defined as a mapping where every element of the domain (set x) is mapped onto exactly one element of the range (set y).
- If every element in the domain is mapped onto exactly one element in the range we say the function is a **one-to-one** function.
- If every element in the domain is mapped onto exactly one element in the range, but some elements in the range arise from more than one element in the domain we say the function is a **many-to-one** function.
- We say that each member of a set is an **element** of that set.
- $x \in \mathbb{R}$ means that x is a member of all the real numbers.

Composite functions

- The composite function **$fg(x)$** means apply g first followed by f .
- The composite function **$gf(x)$** means apply f first followed by g .

Inverse functions

- The **inverse function** of f , written as $f^{-1}(x)$, maps the range back onto the domain.
- The steps for finding the inverse function $f^{-1}(x)$ given $f(x)$ are:
 - Let $f(x) = y$.
 - Change x to y and y to x .
 - Rearrange to get y in terms of x .
 - Write in the correct form.
- The domain of f^{-1} is always the same as the range of f .
- The range of f^{-1} is always the same as the domain of f .
- $ff^{-1}(x)$ and $f^{-1}f(x)$ are always the same as x .
- The graph of $f^{-1}(x)$ is the reflection of the graph of $f(x)$ in the line $y = x$.
- If $f(x) = f^{-1}(x)$, then we say $f(x)$ is a **self-inverse** function. If f is self-inverse, then $ff(x) = x$.

Transformations

- We can transform the graph of a function by moving it horizontally or vertically. This transformation is called a **translation**.
- $f(x + a)$** is a horizontal translation of $-a$.
- $f(x) + a$** is a vertical translation of a .

- We can transform the graph of a function by **reflecting** the graph in one of the axes.
- $-f(x)$** is a reflection in the x -axis.
- $f(-x)$** is a reflection in the y -axis.
- We can transform the graph of a function by **stretching** the graph horizontally or vertically.
- $af(x)$** is a stretch with factor a in the y -direction.
- $f(ax)$** is a stretch with factor $\frac{1}{a}$ in the x -direction.

3) Coordinate geometry

a) Parallel and perpendicular line

Example 1

P has coordinates $(-5, 4)$ and Q has coordinates $(-3, -1)$.

Find

- the gradient of PQ
- the length of PQ
- the coordinates of the mid-point of PQ .

$$\begin{aligned} \text{a) gradient} &= \frac{-1-4}{-3-(-5)} \\ &= \frac{-5}{2} = -\frac{5}{2} \end{aligned}$$

If you start with -1 in the numerator you must start with -3 in the denominator.

$$\begin{aligned} \text{b) length } PQ &= \sqrt{(-3-(-5))^2 + (-1-4)^2} \\ &= \sqrt{(2)^2 + (-5)^2} \\ &= \sqrt{29} = 5.385 \end{aligned}$$

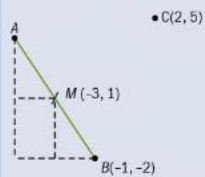
$$\begin{aligned} \text{c) mid-point of } PQ &= \left(\frac{-5+(-3)}{2}, \frac{4+(-1)}{2} \right) \\ &= (-4, 1.5) \end{aligned}$$

Check: -4 is halfway between -3 and -5 .
 1.5 is halfway between 4 and -1 .

Example 2

The mid-point M of the line AB has coordinates $(-3, 1)$. B has coordinates $(-1, -2)$.

Given that a point C has coordinates $(2, 5)$, find the exact length of the line AC .



It is useful to draw a diagram.

Let A be the point (x, y) .

Work out the coordinates of A .

$$\text{Then } \frac{x+(-1)}{2} = -3, \text{ so } x = -5$$

$$\text{Also } \frac{y+(-2)}{2} = 1, \text{ so } y = 4$$

$$A = (-5, 4) \quad C = (2, 5)$$

$$\begin{aligned} \text{length } AC &= \sqrt{(2-(-5))^2 + (5-4)^2} \\ &= \sqrt{1+49} = \sqrt{50} \end{aligned}$$

We are asked for an *exact* answer so do not work out answer as a decimal.

Example 5

Find the gradient of the line that is perpendicular to the lines with these gradients.

Note: Since $m_1 \times m_2 = -1$, $m_2 = -\frac{1}{m_1}$

- a) $\frac{7}{3}$ b) $-\frac{1}{5}$ c) 1 d) -8

a) $-\frac{3}{7}$

Change the sign and invert the fraction.

b) $\frac{5}{1} = 5$

Change the sign and invert the fraction.

c) $-\frac{1}{1} = -1$

$1 = \frac{1}{1}$ and $-8 = -\frac{8}{1}$

d) $\frac{1}{8}$

Example 6

Find in each case whether the lines are parallel to each other, perpendicular to each other, or neither.

- a) $y = 5 - 3x$ b) $x - 2y = 4$ c) $3y = 5x - 2$
 $y = 3x + 4$ $6y = 3x - 1$ $10y = 3 - 6x$

a) $m_1 = -3$ $m_2 = 3$

The lines are not parallel or perpendicular.

The gradient is equal to the coefficient of x .

b) $m_1 = \frac{1}{2}$ $m_2 = \frac{3}{6} = \frac{1}{2}$

The lines are parallel, as $m_1 = m_2$.

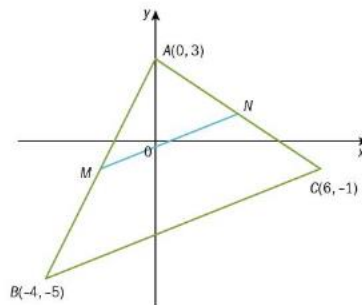
$2y = x - 4$, $y = \frac{1}{2}x - 2$ and $y = \frac{3}{6}x - \frac{1}{6}$

c) $m_1 = \frac{5}{3}$ $m_2 = -\frac{6}{10} = -\frac{3}{5}$

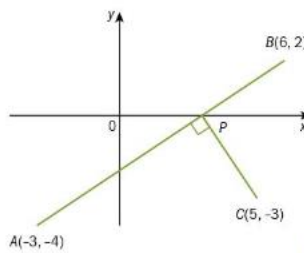
The lines are perpendicular, as $m_1 m_2 = -1$.

$y = \frac{5}{3}x - \frac{2}{3}$ and $y = \frac{3}{10} - \frac{6}{10}x$

3. Find in each case whether the lines are parallel to each other, perpendicular to each other, or neither.
- a) $y = 1 - x$ b) $x - 2y = 4$
 $y = x + 4$ $6y = 3x - 1$
- c) $3y = 9x + 1$ d) $4y = 8x + 1$
 $x + 3y = 4$ $2y = 3 - 4x$
4. The line PQ has a gradient of $\frac{1}{3}$. PQ is parallel to the line joining the points $(3, k)$ and $(-6, 5)$.
Find k .
5. The line joining points $A(x, 3)$ and $B(2, -3)$ is perpendicular to BC where C is the point $(10, 1)$.
Find the value of x .
6. The line that passes through the points $(3, 0)$ and $(-4, y)$ is perpendicular to the line with equation $7x + 4y = 5$. Find the value of y .
7. ABC is a right-angled triangle with AB perpendicular to BC . The coordinates of the points A, B and C are $(-5, 0)$, $(x, -5)$ and $(3, -2)$ respectively. Find the possible values of x .
8. Show that $A(4, 3)$, $B(-3, 2)$, $C(-5, -2)$ and $D(2, -1)$ are the vertices of a parallelogram.
9. The diagram shows triangle ABC . The coordinates of A, B and C are $(0, 3)$, $(-4, -5)$ and $(6, -1)$ respectively. M is the mid-point of AB , and N is the mid-point of AC .
Show that MN is parallel to BC .



10. The diagram shows the straight line AB that cuts the x -axis at P . PC is perpendicular to AB . The point A is $(-3, -4)$, the point B is $(6, 2)$ and the point C is $(5, -3)$.
Show that point P lies on the x -axis and find its coordinates.



b) Equation of straight line

Example 8

Find the equation of the line that passes through the point $(4, -3)$ and is perpendicular to the line with equation $2y = 5x$.

$$2y = 5x \quad m = \frac{5}{2}$$

$$\text{Gradient of the perpendicular line} = -\frac{2}{5}$$

$$\text{Equation: } y + 3 = -\frac{2}{5}(x - 4)$$

$$5y + 15 = -2x + 8$$

$$2x + 5y + 7 = 0$$

← The gradient of the line is equal to the coefficient of x .

← Change the sign and inverse the fraction.

← Multiply $(y + 3)$ by 5 and multiply $(x - 4)$ by -2 .

Example 9

Find the equation of the **perpendicular bisector** of the line PQ , where $P(3, -1)$ and $Q(7, 3)$.

$$\text{Mid-point of } PQ = \left(\frac{3+7}{2}, \frac{-1+3}{2} \right) = (5, 1)$$

$$\text{Gradient } PQ = \frac{3 - (-1)}{7 - 3} = 1$$

$$\text{Gradient of perpendicular} = -1$$

Equation of perpendicular:

$$y - 1 = -1(x - 5)$$

$$x + y - 6 = 0$$

$$m_1 \times m_2 = -1$$

Example 10

$A(4, 3)$, $B(9, 13)$, $C(5, 11)$, and $D(12, 4)$ are four points.

The lines AB and CD meet at P .

Find the equations of AB and CD and hence the coordinates of P .

$$\text{Gradient of } AB = \frac{13-3}{9-4} = 2$$

$$\begin{aligned} \text{Equation } AB: y - 3 &= 2(x - 4) \\ y &= 2x - 5 \end{aligned}$$

← Use the coordinates of A or B .

$$\text{Gradient of } CD = \frac{4-11}{12-5} = -1$$

$$\begin{aligned} \text{Equation } CD: y - 11 &= -1(x - 5) \\ y &= -x + 16 \end{aligned}$$

← Use the coordinates of C or D .

$$\text{At } P: 2x - 5 = -x + 16$$

$$3x = 21$$

$$x = 7$$

← Both equal y .

$$\text{When } x = 7, y = 2(7) - 5 = 9$$

$$\text{Coordinates of } P = (7, 9)$$

← Use equation of AB or CD .

2. Find the equation of the line AB with the given pairs of coordinates.

- a) $A(1, 5)$ and $B(3, 7)$ b) $A(-1, -2)$ and $B(-3, 4)$
 c) $A(5, -3)$ and $B(-2, 4)$ d) $A(-2, 4)$ and $B(-6, 0)$
 e) $A(-8, -1)$ and $B(2, -6)$ f) $A(9, -3)$ and $B(-6, -2)$

3. Find the equation of the line parallel to the line $y = 4 - 3x$ and passing through the point $(5, 7)$.

4. Find the equation of the line perpendicular to the line $y = 5x$ and passing through the point $(2, -1)$.

5. Find the equation of the line through the point $(-1, 3)$ parallel to the line with equation $2x - 5y = 10$.

6. Find the equation of the line through the point $(-1, 3)$ perpendicular to the line with equation $6x + 9y - 7 = 0$.

7. Find the equation of the line through the point $(1, 8)$ perpendicular to the line with equation $3x - 4y + 4 = 0$.

8. A is the point $(3, -4)$ and B is the point $(-5, -10)$. Find the equation of the perpendicular bisector of AB .

9. $A(9, 0)$, $B(5, 8)$, $C(1, 1)$ and $D(3, 2)$ are four points.

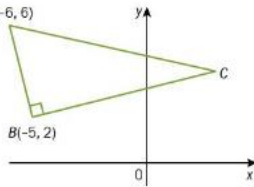
The lines AB and CD meet at P .

Find the equations of AB and CD and hence the coordinates of P .

10. $A(1, -2)$, $B(-1, -3)$, $C(-2, 14)$, and $D(2, 2)$ are four points.

The lines AB and CD meet at P . Find the coordinates of P .

11. $A(-6, 6)$



The diagram shows triangle ABC with AB perpendicular to BC . A is the point $(-6, 6)$ and B is the point $(-5, 2)$. The gradient of AC is $-\frac{2}{9}$. Find the coordinates of C .

12. The straight line with equation $3x - 2y = 6$ cuts the x -axis at P and the y -axis at Q .

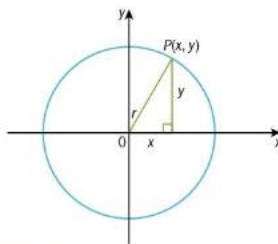
a) Find the coordinates of P and Q .

b) Find the equation of the line that is perpendicular to PQ and passes through the point $(4, 3)$.

13. $A(5, -5)$, $B(3, -7)$, $C(12, 3)$ and $D(-3, -6)$ are four points.

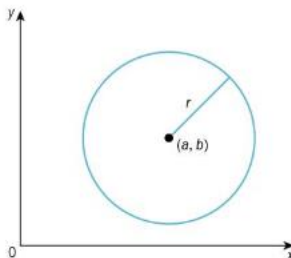
c) Circles

If we take any point $P(x, y)$ on the circumference of a circle, centre O , we can work out the equation of the circle using Pythagoras' theorem.



The equation of a circle, centre $(0, 0)$, radius r , is given by $x^2 + y^2 = r^2$.

If we translate this circle by the vector $\begin{pmatrix} a \\ b \end{pmatrix}$, we can get the equation of any circle, centre (a, b) , radius r .



The equation of a circle, centre (a, b) , radius r is given by $(x - a)^2 + (y - b)^2 = r^2$.

If we expand this equation, we get $x^2 - 2ax + a^2 + y^2 - 2by + b^2 = r^2$. When this is simplified, we get the expanded form of the equation of a circle:

$$x^2 + y^2 - 2ax - 2by + c = 0 \quad \text{where } c = a^2 + b^2 - r^2$$

Note: When the circle is given in this expanded form, we can use the method of completing the square to write the equation in the form $(x - a)^2 + (y - b)^2 = r^2$.

The expanded form of the equation of a circle is given by

$$x^2 + y^2 + 2gx + 2fy + c = 0, \text{ where } c = f^2 + g^2 - r^2.$$

Example 12

Find the centre and radius of the circle with equation $x^2 + y^2 - 10x + 12y + 12 = 0$.

$$x^2 - 10x + y^2 + 12y + 12 = 0$$

$$(x - 5)^2 - 25 + (y + 6)^2 - 36 + 12 = 0$$

$$(x - 5)^2 + (y + 6)^2 = 49$$

Centre is $(5, -6)$, radius = 7.

First rearrange the equation to this form.

Complete the square for x and y .

Rearrange to this form.

$$a = 5, b = -6, r = \sqrt{49}$$

Example 13

Find the equation of the circle with centre $(4, -3)$ that passes through the point $(-2, 5)$.

$$(x - 4)^2 + (y + 3)^2 = r^2$$

$$(-2 - 4)^2 + (5 + 3)^2 = r^2$$

$$36 + 64 = r^2$$

$$r^2 = 100$$

Equation of the circle is

$$(x - 4)^2 + (y + 3)^2 = 100$$

Write the equation of circle centre $(4, -3)$.

Substitute for $x = -2$ and $y = 5$.

Simplify

Work out r^2 .

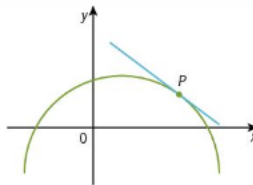
Write down the equation of the circle.

d) Points of intersection

When a line touches a curve in exactly one place we call that line a **tangent** to the curve.

This also means that there will only be one solution when solving the equations simultaneously.

If you form a quadratic equation there will be only **one (repeated) root**,
i.e. $b^2 - 4ac = 0$



Example 14

Find the set of values of k for which the line $x = y + 2$ intersects the curve $y = x^2 + 3x + k$ at two distinct points.

They meet when $x^2 + 3x + k = x - 2$.

$$x^2 + 2x + (k + 2) = 0$$

Rearrange as a quadratic equation.

There are two distinct roots when $b^2 - 4ac > 0$.

$$(2)^2 - 4(1)(k + 2) > 0$$

$a = 1, b = 2$ and $c = k + 2$

$$-4k - 4 > 0$$

$$-4k > 4$$

Use inequality signs throughout.

$$k < -1$$

Example 15

A line has equation $y = 4k - x$ and a curve has equation $y = 3x - kx^2$, where k is a constant.

- a) Find the two values of k for which the line is a tangent to the curve.
b) Hence find the coordinates of the points where the line touches the curve.

a) The line and the curve meet when

$$4k - x = 3x - kx^2$$

Both equal y .

$$kx^2 - 4x + 4k = 0$$

Rearrange in the form $ax^2 + bx + c = 0$.

The line is a tangent so there is one repeated root.

$$(-4)^2 - 4(k)(4k) = 0$$

$$b^2 - 4ac = 0$$

$$16 = 16k^2, k^2 = 1$$

$$k = 1 \text{ or } k = -1$$

b) When $k = 1$:

$$x^2 - 4x + 4 = 0$$

Substitute in $kx^2 - 4x + 4k = 0$.

$$(x - 2)(x - 2) = 0$$

$$x = 2, y = 4k - x = 4 - 2 = 2$$

When $k = -1$:

$$-x^2 - 4x - 4 = 0$$

$$x^2 + 4x + 4 = 0$$

$$(x + 2)(x + 2) = 0$$

$$x = -2, y = 4k - x = -4 + 2 = -2$$

Line meets curve at $(2, 2)$ and $(-2, -2)$.

The question asked for coordinates.

Example 16

Determine the equation of the tangent to the circle, centre C , with equation $x^2 + y^2 - 2x - 4y - 5 = 0$ at the point $A(2, 5)$. Give your answer in the form $ax + by + c = 0$, where a , b and c are integers.

$$(x - 1)^2 - 1 + (y - 2)^2 - 4 - 5 = 0$$

Complete the square to find the centre.

C is $(1, 2)$.

We do not need to find the radius.

$$\text{Gradient of } AC = \frac{5-2}{2-1} = 3$$

Use the points $(2, 5)$ and $(1, 2)$.

$$\text{Gradient of the tangent} = -\frac{1}{3}$$

The tangent is perpendicular to the radius.

$$\text{Equation of the tangent: } y - 5 = -\frac{1}{3}(x - 2)$$

The gradient is $-\frac{1}{3}$ through the point $(2, 5)$.

$$3y - 15 = -x + 2$$

$$x + 3y - 17 = 0$$

Simplify and put in the required form.

Example 17

Determine the shortest distance from the point $P(1, 1)$ to the circle, centre C , with equation $x^2 + y^2 + 6x - 8y + 21 = 0$.

$$(x + 3)^2 - 9 + (y - 4)^2 - 16 + 21 = 0$$

Complete the square to find C and radius r .

C is $(-3, 4)$, $r = \sqrt{4} = 2$.

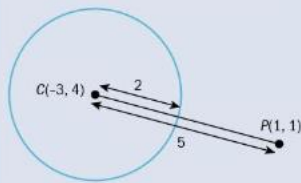
Find the centre and the radius.

$$PC = \sqrt{(1 - (-3))^2 + (1 - 4)^2} = 5$$

Find the distance between P and C .

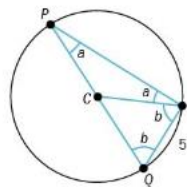
$$\text{Shortest distance} = 5 - 2 = 3$$

See diagram below.



Remember that the tangent to a point on a circle is always perpendicular to the radius at that point and that angles in a semicircle are always right angles. The diagram reminds you why $\angle PRQ = 90^\circ$ if PQ is a diameter of a circle with centre C . Drawing the radius CR creates two isosceles triangles, with one base angle from each making up $\angle PRQ$.

You can be asked to use these elementary geometrical properties of circles when working with coordinate geometry.

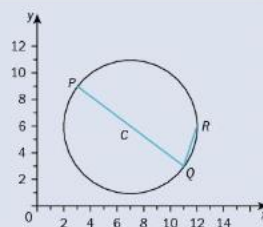


Coordinate geometry

Example 18

The diagram shows a circle, centre C , with equation $(x - 7)^2 + (y - 6)^2 = 25$. PQ is a diameter, where P is $(3, 9)$ and Q is $(11, 3)$. $R(12, 6)$ lies on the circle.

Show that the line perpendicular to QR and passing through R goes through P .



QR has gradient $\frac{6-3}{12-11} = 3$, so the gradient

of the perpendicular is $-\frac{1}{3}$.

Gradients of perpendicular lines multiply to -1 .

The equation of the perpendicular line through $(12, 6)$ is given by

$$y - 6 = -\frac{1}{3}(x - 12)$$

Substitute the gradient and known point to get the equation of the line.

$$y = -\frac{1}{3}x + 10$$

Rearrange to standard form.

When $x = 3$, $y = -\frac{1}{3} \times 3 + 10 = 9$

so $P(3, 9)$ lies on the line.

Always finish the answer to a 'show that' question with a statement.

Example 19

AB is the diameter of a circle, where A is $(2, 6)$ and B is $(8, 2)$.

The tangent to the circle at B meets the x -axis at P and the y -axis at Q . Find the coordinates of P and Q .

A tangent is perpendicular to the diameter and we have two points defining the diameter.

AB has gradient $\frac{2-6}{8-2} = -\frac{2}{3}$, so the

gradient of the tangent at B is $\frac{3}{2}$.

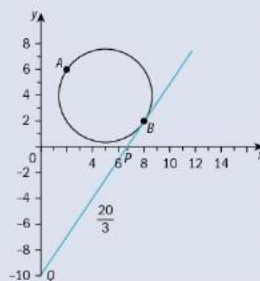
The equation of the perpendicular line through $(8, 2)$ is given by

$$y - 2 = \frac{3}{2}(x - 8)$$

Substitute the gradient and known point to get the equation of the line.

$$y = \frac{3}{2}x - 10$$

Rearrange to standard form.



Continued on the next page

When $y = 0$, $0 = \frac{3}{2}x - 10 \Rightarrow x = \frac{20}{3}$

Substitute $x = 0$ and $y = 0$ to find coordinates of intercepts.

so P is $\left(\frac{20}{3}, 0\right)$.

When $x = 0$, $y = -10$ so Q is $(0, -10)$.

Example 20

The points P , Q and R have coordinates $(0, 2)$, $(4, 0)$ and $(8, 8)$ respectively.

Use Pythagoras' theorem to show that PQR is a right-angled triangle and hence find the equation of the circle that passes through P , Q and R .

$PQ^2 = 2^2 + 4^2 = 20$ $QR^2 = 4^2 + 8^2 = 80$

Calculate the squares of sides of triangle.

$PR^2 = 8^2 + 6^2 = 100$

so $PQ^2 + QR^2 = PR^2$ and PQR is a right-angled triangle with PR as the hypotenuse with the right angle at Q .

Use Pythagoras' theorem as required.

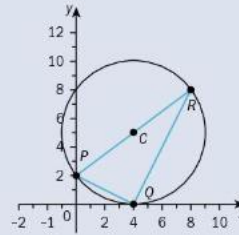
The circle through P , Q and R therefore has PR as diameter.

If C is the centre of the circle, then C is the mid-point of PR and the radius of the circle is half PR .

C is $(4, 5)$, radius = 5.

The equation of the circle is $(x - 4)^2 + (y - 5)^2 = 25$.

Substitute centre coordinates and radius to get the equation of the circle.



Summary:

Mid-point of a line segment

- The mid-point of the line joining (x_1, y_1) and (x_2, y_2) is given by $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$.

Gradient of a line

- The gradient of the line joining (x_1, y_1) and (x_2, y_2) is given by $\frac{y_2 - y_1}{x_2 - x_1}$.

Length of a line segment

- The length of the line segment between (x_1, y_1) and (x_2, y_2) is given by $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Parallel and perpendicular lines

- Two straight lines are parallel if their gradients are the same ($m_1 = m_2$).
- Two straight lines are perpendicular if the product of their gradients = -1 ($m_1 \times m_2 = -1$).

Equation of a line

- The equation of the straight line through the point (x_1, y_1) and with gradient m is:
 $(y - y_1) = m(x - x_1)$ (given point and gradient).
- The equation of the straight line through the points (x_1, y_1) and (x_2, y_2) is:
 $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$ (given two points).

Equation of a circle

- The equation of a circle, centre (a, b) , radius r , is given by $(x - a)^2 + (y - b)^2 = r^2$.
- The expanded form of the equation of a circle is given by $x^2 + y^2 + 2gx + 2fy + c = 0$,
where $c = f^2 + g^2 - r^2$.

Tangent to a curve

- When a line touches a curve in exactly one place we call that line a tangent to the curve.
- When you form a quadratic equation in order to solve the equation of the tangent and the equation of the quadratic simultaneously, there will be only one repeated root, i.e. use $b^2 - 4ac = 0$.

4) Circular measure

a) Radians

Example 1

Convert 165° to radians.

$$165 \times \frac{\pi}{180} = \frac{11\pi}{12} = 2.88 \text{ radians (3 s.f.)}$$

Multiply by $\frac{\pi}{180}$.

Example 2

Convert $\frac{2\pi}{3}$ radians to degrees.

$$\frac{2\pi}{3} \times \frac{180}{\pi} = 120^\circ$$

Multiply by $\frac{180}{\pi}$.

Or

$$\frac{2\pi}{3} = 2 \times \frac{\pi}{3} = 2 \times 60^\circ$$

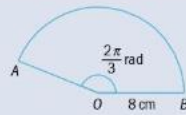
Note: It is a useful skill to be able to find multiples of known equivalents quickly in your head.

b) Arc length and sector

Example 3

OAB is a sector of a circle with centre O and radius 8 cm.

- Find the area of the sector in terms of π .
- Find the perimeter of the sector in terms of π .



$$\text{a) } A = \frac{1}{2}r^2\theta = \frac{1}{2} \times 8^2 \times \frac{2\pi}{3} = \frac{64\pi}{3} \text{ cm}^2$$

$$\text{b) } s = r\theta = 8 \times \frac{2\pi}{3} = \frac{16\pi}{3}$$

$$\text{Perimeter} = \text{arc length} + 8 + 8 = \frac{16\pi}{3} + 16 \text{ cm}$$

Substitute $r = 8$ and $\theta = \frac{2\pi}{3}$ into the formulae.

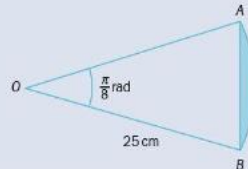
Example 4

OAB is a sector of a circle with centre O and radius 25 cm.

The line AB is a **chord** of the circle.

Find the area of the shaded region.

(You are given that the area of triangle OAB may be found by using the formula $\frac{1}{2} \times OA \times OB \times \sin AOB$.)



$$\text{Area of shaded region} = \text{Area of sector } OAB - \text{Area of triangle } OAB$$

$$\text{Area of sector } OAB = \frac{1}{2}r^2\theta = \frac{1}{2} \times 25^2 \times \frac{\pi}{8}$$

Find the area of the sector.

$$\text{Area of triangle } OAB = \frac{1}{2} \times 25 \times 25 \times \sin \frac{\pi}{8}$$

Find the area of the triangle.

$$\text{Area of the shaded region} = \frac{1}{2} \times 25^2 \times \frac{\pi}{8} - \frac{1}{2} \times 25^2 \times \sin \frac{\pi}{8}$$

$$= 3.13 \text{ cm}^2 \text{ (3 s.f.)}$$



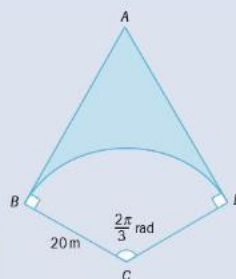
You should use the radian mode in your scientific calculator.

Example 5

In the diagram, BD is the arc of a circle with centre C and radius 20 m .

Angle $BCD = \frac{2\pi}{3}$ radians.

AB and AD are tangents to the circle at B and D , respectively.



a) Show that $AB = 20\sqrt{3}\text{ cm}$.

b) Show that the area of the shaded region is $400\left(\sqrt{3} - \frac{\pi}{3}\right)\text{ cm}^2$.

c) Find the perimeter of the shaded region in terms of π and $\sqrt{3}$.

Note: You may assume that $\tan \frac{\pi}{3} = \sqrt{3}$ in this question. Exact trigonometric ratios will be studied in Chapter 5.

a) Angle $BCA = \frac{\pi}{3}$

$$\tan \frac{\pi}{3} = \frac{AB}{20}$$

$$AB = 20 \tan \frac{\pi}{3}$$

$$AB = 20\sqrt{3}\text{ cm}$$

Use the right-angled triangle ABC .

Find AB .

b) Shaded area = $2 \times$ area of triangle ABC – area of sector DCB

$$= 2 \times \frac{1}{2} \times 20\sqrt{3} \times 20 - \frac{1}{2} \times 20^2 \times \frac{2\pi}{3}$$

$$= 400\sqrt{3} - \frac{400\pi}{3}$$

$$= 400\left(\sqrt{3} - \frac{\pi}{3}\right)\text{ cm}^2$$

Use the formulae: area of a triangle = $\frac{1}{2}bh$
and area of a sector = $\frac{1}{2}r^2\theta$.

c) Perimeter = $AB +$ arc length $BD + AD$

$$= 20\sqrt{3} + 20 \times \frac{2\pi}{3} + 20\sqrt{3}$$

$$= 40\sqrt{3} + \frac{40\pi}{3}\text{ cm}$$

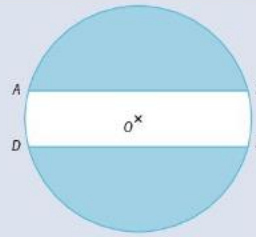
Use arc length = $r\theta$ and $AD = AB$.

Leave your answer in exact form.

Example 6

The diagram shows a shape $ABCD$ made by removing the two shaded regions from the circle of radius 8 cm and with centre O .

AB is parallel to DC . $AB = DC = 14$ cm.



- Find angle AOB . Give your answer in radians correct to 4 significant figures.
- Find the perimeter of $ABCD$.
- Find the area of $ABCD$.

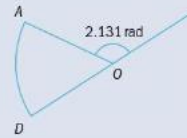
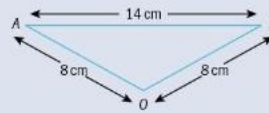
- a) Using the cosine rule:

$$\begin{aligned}\cos AOB &= \frac{8^2 + 8^2 - 14^2}{2 \times 8 \times 8} \\ &= \frac{-17}{32}\end{aligned}$$

Recall the cosine rule:

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{Angle } AOB = 2.131 \text{ rad (4 s.f.)}$$



- b) Perimeter of $ABCD = 2 \times \text{arc length } AD + 2 \times 14$

$$\text{Angle } AOD = \pi - 2.131$$

$$= 1.011 \text{ rad}$$

$$\text{Perimeter} = (2 \times 8 \times 1.011) + (2 \times 14)$$

$$= 44.2 \text{ cm (3 s.f.)}$$

Give your answer to 3 significant figures.

- c) Area of triangle $AOB = \frac{1}{2} \times 8 \times 8 \times \sin 2.131 = 27.11$

$$\text{Area of sector } OAD = \frac{1}{2} \times 8^2 \times 1.011 = 32.35$$

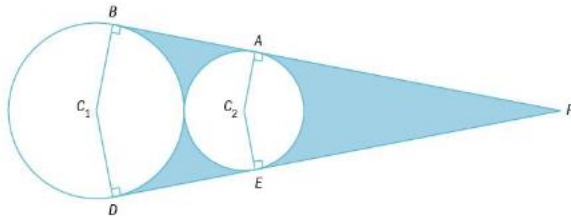
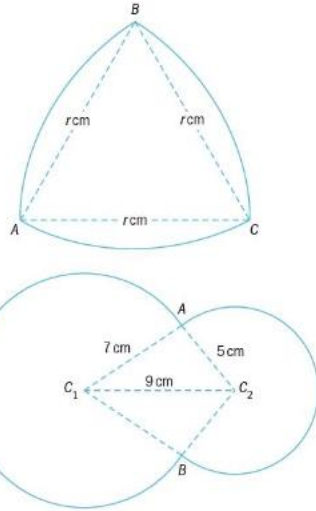
$$\text{Area of } ABCD = 2 \times 27.11 + 2 \times 32.35$$

$$= 119 \text{ cm}^2 \text{ (3 s.f.)}$$

Use $\frac{1}{2}ab \sin C$ and $\frac{1}{2}r^2\theta$.

2 × area of triangle + 2 × area of sector

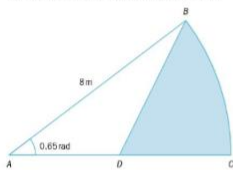
4. The diagram shows a shape ABC .
The boundary of the shape is made from three arcs.
The arcs AB , BC and AC have centres C , A , B , respectively.
- a) Find the perimeter of the shape in terms of r and π .
b) Find the area of the shape when $r = 15$ cm.
5. A metal plate is made from two circles.
One circle is of radius 7 cm.
The other circle is of radius 5 cm.
Their centres C_1 and C_2 are 9 cm apart.
- a) Find, in radians, angles AC_1B and AC_2B .
b) Find the area of the metal plate.
c) Find the perimeter of the metal plate.
6. In the diagram below, PAB and PED are tangents to two circles which touch and have centres C_1 and C_2 , respectively and radii 10 cm and 6 cm, respectively.



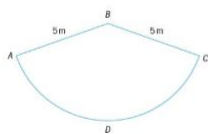
Summary exercise 4

EXAM-STYLE QUESTIONS

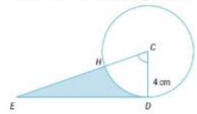
1. In the diagram, ABC is a sector of a circle with centre A and radius 8 m. The size of angle BAC is 0.65 rad and D is the mid-point of AC . Find
- a) the length of the arc BC
b) the area of the shaded region BCD .



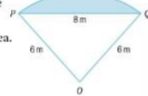
2. In the diagram below, $ABCD$ is a sector of a circle with centre B and radius 5 m. Given that the length of the arc ADC is $\frac{15\pi}{4}$ m, find
- a) the exact size of angle ABC in radians
b) the exact area of the sector $ABCD$.



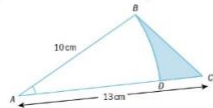
3. A circle has centre C and radius 4 cm. The diagram shows the circle with a point D which lies on the circle. The tangent at D passes through the point E . $EC = \sqrt{65}$ cm. Find
- a) the size in radians of angle DCE
b) the area of the shaded region DEH .



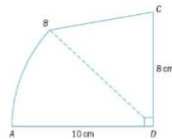
4. In the diagram below, OPQ is a sector of a circle, radius 6 m. The straight line PQ is 8 m long. Calculate
- a) the area of the sector OPQ
b) the shaded area.



5. The diagram below shows a triangle ABC with $AB = 10$ cm, $AC = 13$ cm and angle $BAC = 0.5$ rad. BD is an arc of a circle with centre A and radius 10 cm. Find
- a) the length of arc BD
b) the perimeter of the shaded region
c) the area of the shaded region.

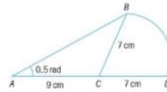


6. The diagram shows a shape $ABCD$. The curve AB is an arc of a circle with centre D . Angle ADB is 0.75 rad. Find
- a) the length of the arc AB
b) the area of the sector ADB
c) the area of $ABCD$.

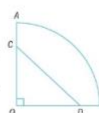


EXAM-STYLE QUESTION

7. A shape is made from joining a triangle ABC to a sector CBD of a circle with radius 7 cm and centre C . The points A , C and D lie on a straight line with $AC = 9$ cm. Angle $BAC = 0.5$ radians. Find
- a) the angle ACB giving your answer in radians
b) the area of $ABCD$.

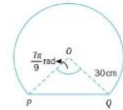


8. The diagram shows the sector OAB of a circle with centre O and radius 20 m. Angle $AOB = \frac{\pi}{2}$ radians. The straight line CD divides the sector into two regions which have equal areas and the lengths of OC and OD are equal. Calculate the exact length of OC .



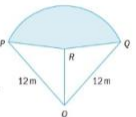
EXAM-STYLE QUESTION

9. The diagram shows a shape made from the arc of a circle with centre O and radius 30 cm and the straight line PQ . The size of angle POQ is $\frac{7\pi}{9}$ radians.

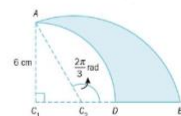


Find the total perimeter of the shape.

10. The diagram shows a sector OPQ of a circle with centre O and radius 12 m. $PR = QR = OR = 8$ m. Work out the area of the shaded region PQR .



11. AD is an arc of a circle with centre C_1 and radius 6 cm. AB is an arc of a circle with centre C_2 . The size of angle AC_2D is $\frac{2\pi}{3}$ radians.



Calculate

- a) the radius of the circle with centre C_2
b) the area of the shaded region ABD .

Summary:

Radians and degrees

- $360^\circ = 2\pi$ radians.
- To convert from degrees to radians, multiply by $\frac{\pi}{180}$.
- To convert from radians to degrees, multiply by $\frac{180}{\pi}$.

Degrees	0°	30°	45°	60°	90°	180°	270°	360°
Radians	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

Arc length and area of a sector

- The angle for a whole circle = 2π radians.
- Arc length = $r\theta$ (θ measured in radians).
- Area of sector = $\frac{1}{2}r^2\theta$ (where θ is measured in radians and r is the radius of the circle).

5) Trigonometry

a) Exact values

The unit circle can be used to define the trigonometric ratios.

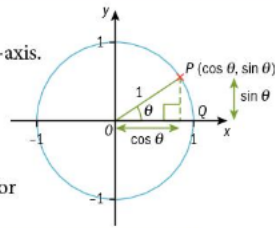
The angle θ is measured in an anticlockwise direction from the positive x -axis.

$\cos \theta$ can be defined as the x coordinate of P .

$\sin \theta$ can be defined as the y coordinate of P .

$\tan \theta$ can be defined as $\frac{\sin \theta}{\cos \theta}$.

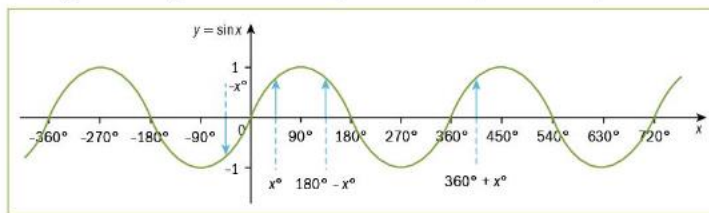
The diagram can be used to write down the values of $\sin$, $\cos$ and $\tan$ for 0° , 90° , 180° , 270° and 360° .



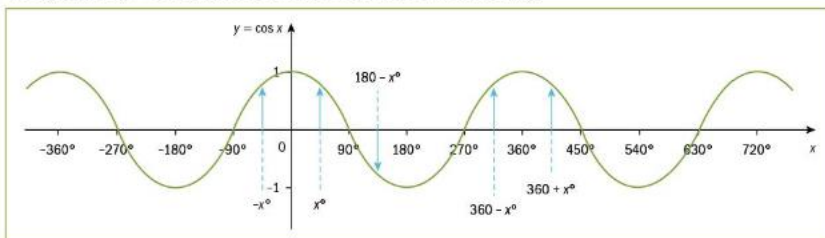
Note:

b) Graph of trigonometric functions

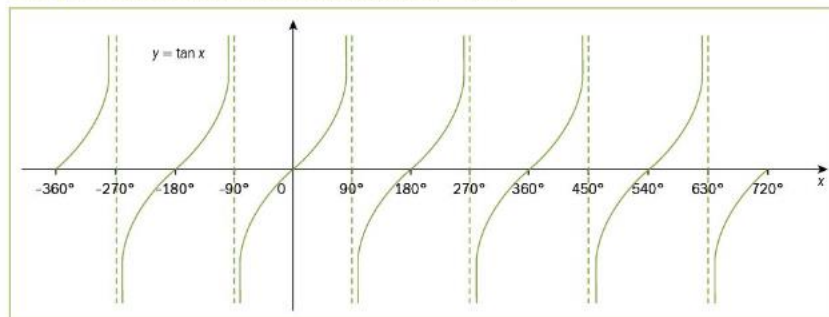
We say $y = \sin x$ is periodic and has a **period** of 360° (or 2π radians).



Periodic



$\cos(-x) = \cos x$



Example 3

Find all the angles x (to the nearest degree), where $0^\circ < x < 360^\circ$, such that $\sin x = 0.88$.

$$x = \sin^{-1}(0.88) \\ = 62^\circ$$

Use a calculator to find the principal value.

$$x = 62^\circ, 180^\circ - 62^\circ$$

Use a graph sketch to find the second angle in the stated interval.

$$x = 62^\circ, 118^\circ$$

Example 4

Solve $\tan x = 2.35$ for $0^\circ < x < 720^\circ$.

Give your answers correct to one decimal place.

$$x = \tan^{-1}(2.35) \\ = 66.9^\circ \text{ (1 d.p.)}$$

This is the principal value given by a calculator.

$$x = 66.9^\circ, 180^\circ + 66.9^\circ, 360^\circ + 66.9^\circ, 540^\circ + 66.9^\circ$$

$$x = 66.9^\circ, 246.9^\circ, 426.9^\circ, 606.9^\circ \text{ (1 d.p.)}$$

Add on multiples of 180° to give all values in the interval $0^\circ < x < 720^\circ$.

1. Find the period of each of these functions.

a) $y = \sin 2x$

b) $y = \cos 5x$

c) $y = \cos x + 1$

d) $y = 5 \sin(x - 30)^\circ$

e) $y = \tan \frac{1}{2}x$

f) $y = 2 \tan 3x - 4$

g) $y = \tan(x + 45)^\circ$

h) $y = 3 \sin(2x - 60)^\circ$

2. Sketch, on separate diagrams, the graphs of

a) $y = \sin 3x$ for $0^\circ \leq x \leq 360^\circ$

b) $y = -\cos 2x$ for $0^\circ \leq x \leq 360^\circ$

c) $y = \tan \frac{1}{2}x$ for $-360^\circ \leq x \leq 360^\circ$

d) $y = 2 \sin 4x$ for $0^\circ \leq x \leq 180^\circ$

e) $y = 3 \tan(x + 30^\circ)$ for $0^\circ \leq x \leq 360^\circ$

f) $y = 3 \sin 2x - 1$ for $0^\circ \leq x \leq 360^\circ$.

c) Trigonometric equations

Example 8

Solve the equation $4 \sin x + 3 = 5$ for $0^\circ \leq x \leq 360^\circ$.

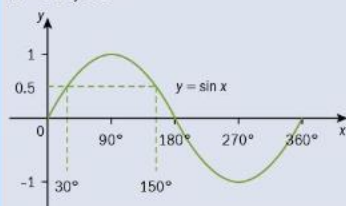
$$4 \sin x + 3 = 5$$

$$4 \sin x = 2$$

$$\sin x = \frac{1}{2}$$

$$x = 30^\circ, 180^\circ - 30^\circ$$

$$x = 30^\circ, 150^\circ$$



First find the value of $\sin x$.

Use a sketch graph to help you find all the angles in the required interval.

Example 9

Solve the equation $3 \cos 2\theta = -1$ for $0^\circ \leq \theta \leq 360^\circ$.

Values for $0^\circ \leq \theta \leq 360^\circ$ are required, so values for $0^\circ \leq 2\theta \leq 720^\circ$ will be needed.

$$3 \cos 2\theta = -1$$

$$\cos 2\theta = -\frac{1}{3}$$

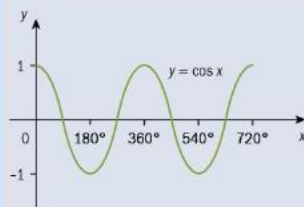
$$2\theta = 109.47\dots^\circ, \quad 360^\circ - 109.47\dots^\circ$$

$$360^\circ + 109.47\dots^\circ, \quad 720^\circ - 109.47\dots^\circ$$

$$2\theta = 109.47\dots^\circ, \quad 250.52\dots^\circ$$

$$469.47\dots^\circ, \quad 610.52\dots^\circ$$

$$\theta = 54.7^\circ, 125.3^\circ, 234.7^\circ, 305.3^\circ$$



Use your calculator to find the principal value $109.47\dots^\circ$.

Use the graph to find the other angles.

Note: You should normally give angles in degrees correct to 1 decimal place, so make sure you work with more than this to avoid rounding errors. Remember to work out all the values from the graph before you divide by 2.

1. Solve these equations for $0^\circ \leq x \leq 360^\circ$.

a) $4 \sin x = 3$	b) $5 \tan x - 1 = 9$	c) $6 \cos x + 5 = 8$
d) $8 \sin(x + 20^\circ) = 5$	e) $10 - 3 \cos x = 9$	f) $9 \sin(x - 15^\circ) = -4$
2. In the following equations, find all the values of θ between 0° and 360° .

a) $\cos \theta = 0.776$	b) $\sin 2\theta = -0.364$	c) $\tan 3\theta = 1.988$
d) $\cos \frac{\theta}{2} = -0.379$	e) $\tan \frac{\theta}{2} = -1.030$	f) $\sin \frac{3\theta}{2} = 0.664$
3. Solve the following equations for $0 \leq x \leq 2\pi$.

a) $3 \tan x + 2 = 5$	b) $\sin\left(x - \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}$	c) $\sin 2x = \frac{1}{2}$
d) $6 \cos 3x - 3 = 0$	e) $7 - 2 \tan 4x = 13$	f) $10 \cos \frac{x}{3} = 1$
4. Solve $6 \cos 30x - 3 = 0$ for $0^\circ \leq x < 24^\circ$.
5. Solve $3 \tan x = \sqrt{27}$ for $-180^\circ \leq x < 180^\circ$.
6. Solve $5 + 2 \cos\left(3\theta - \frac{\pi}{4}\right) = 6$ for $-\pi \leq \theta < \pi$.

d) Trigonometric identities

Example 10Simplify a) $\frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta}$ b) $\frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta}$.

$$\text{a) } \frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\cos \theta} = \frac{\sin \theta (\sin^2 \theta + \cos^2 \theta)}{\cos \theta} \quad \leftarrow \text{Factorise the numerator.}$$

$$= \frac{\sin \theta}{\cos \theta} \quad \leftarrow \text{Use } \sin^2 \theta + \cos^2 \theta = 1.$$

$$= \tan \theta \quad \leftarrow \text{Use } \frac{\sin \theta}{\cos \theta} = \tan \theta.$$

$$\text{b) } \frac{1 + \cos \theta}{\sin \theta} + \frac{\sin \theta}{1 + \cos \theta} = \frac{(1 + \cos \theta)^2 + \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \quad \leftarrow \text{Add the fractions.}$$

$$= \frac{1 + 2 \cos \theta + \cos^2 \theta + \sin^2 \theta}{\sin \theta (1 + \cos \theta)} \quad \leftarrow \text{Multiply out } (1 + \cos \theta)^2.$$

$$= \frac{1 + 2 \cos \theta + 1}{\sin \theta (1 + \cos \theta)} \quad \leftarrow \sin^2 \theta + \cos^2 \theta = 1$$

$$= \frac{2(1 + \cos \theta)}{\sin \theta (1 + \cos \theta)} \quad \leftarrow \text{Factorise the numerator.}$$

$$= \frac{2}{\sin \theta} \quad \leftarrow \text{Divide the numerator and denominator by } (1 + \cos \theta), \text{ for } \cos \theta \neq -1.$$

Example 11Prove the identity $\frac{\sin \theta}{\tan \theta (1 - \cos \theta)} = \frac{\cos \theta}{1 - \cos \theta}$, $\sin \theta \neq 0$.

$$\frac{\sin \theta}{\tan \theta (1 - \cos \theta)} = \frac{\sin \theta}{\frac{\sin \theta}{\cos \theta} (1 - \cos \theta)} \quad \leftarrow \text{Start with the left hand side and replace } \tan \theta \text{ with } \frac{\sin \theta}{\cos \theta}.$$

$$= \frac{\cancel{\sin \theta}^1 \cos \theta}{\cancel{\sin \theta}_1 (1 - \cos \theta)} \quad \leftarrow \div \frac{\sin \theta}{\cos \theta} \text{ is the same as } \times \frac{\cos \theta}{\sin \theta}.$$

$$= \frac{\cos \theta}{1 - \cos \theta} \quad \leftarrow \text{Divide numerator and denominator by the common factor } \sin \theta.$$

3. Simplify

a) $\frac{\sin^3 \theta + \sin \theta \cos^2 \theta}{\sin \theta}$ b) $1 + \sin \theta + \frac{\cos^2 \theta}{1 + \sin \theta}$.

4. Prove the identity $\frac{\sin^4 \theta}{\tan \theta} = \sin \theta \cos \theta - \sin \theta \cos^3 \theta$.

5. Prove the following identities

a) $\cos^3 \theta = \cos \theta - 2 \sin^2 \theta \cos \theta + \cos \theta \sin^4 \theta$

b) $(4 \sin \theta + 3 \cos \theta)^2 + (3 \sin \theta - 4 \cos \theta)^2 = 25$

c) $\tan^2 A - \tan^2 B = \frac{\sin^2 A - \sin^2 B}{\cos^2 A \cos^2 B}$.

6. **Show that** $(1 + \sin \theta + \cos \theta)^2 = 2(1 + \sin \theta)(1 + \cos \theta)$.

7. Simplify

a) $\frac{\sin^2 \theta - \cos^2 \theta}{\sin \theta - \cos \theta}$ b) $\frac{\sin^4 \theta - \cos^4 \theta}{\sin^2 \theta - \cos^2 \theta}$ c) $\frac{\sin^4 \theta - \cos^4 \theta}{\sin \theta - \cos \theta}$.

8. Prove the following identities.

a) $\tan \theta = \sqrt{\frac{\sin^2 \theta}{1 - \sin^2 \theta}}$ b) $\frac{1 - \sin^2 \theta}{\tan \theta} = \frac{\cos^3 \theta}{\sin \theta}$.

c) $\frac{\sin^2 \theta}{1 - \cos \theta} = 1 + \cos \theta$ d) $(2 - \cos^2 \theta)^2 - 4 \sin^2 \theta = \cos^4 \theta$

9. a) Express $2 \cos x - \sin^2 x$ in the form $(\cos x + p)^2 + q$, stating the values of p and q .

b) Use your answer to part (a) to find the maximum and minimum values of $2 \cos x - \sin^2 x$.

Solve the equations in questions 1 to 8 for $0^\circ \leq \theta \leq 360^\circ$.

1. $\cos^2 \theta - \sin^2 \theta = 0$.

2. $2 \cos^2 \theta + \sin \theta = 1$.

3. $3 \sin^2 \theta = \cos^2 \theta$.

4. $\cos^2 \theta - 4 \sin^2 \theta = 1$.

5. $1 + \sin \theta \cos^2 \theta = \sin \theta$

6. $6 \sin(\theta + 70)^\circ - 5 \cos(\theta + 70)^\circ = 0$

7. $3 \cos 2\theta = 4 \sin 2\theta$.

8. $2 \sin \theta = 3 \tan \theta$.

9. Solve the equation $\sin \theta + 5 \cos \theta = 3 \sin \theta$ for $0 \leq \theta \leq 2\pi$.

10. Solve the equation $2 \cos \theta - 5 \sin \theta = 4 \cos \theta + 3 \sin \theta$ for $0 \leq \theta \leq 2\pi$.

11. a) Simplify $\frac{6 - 6 \cos^2 \theta}{2 \sin \theta}$.

b) Hence solve the equation $6 - 6 \cos^2 \theta = 3 \sin \theta$ for $0 \leq \theta \leq 2\pi$.

12. Solve the equation $\tan^3 \theta - \tan^2 \theta - 2 \tan \theta = 0$ for $0^\circ \leq \theta \leq 180^\circ$.

Summary:

Chapter Summary

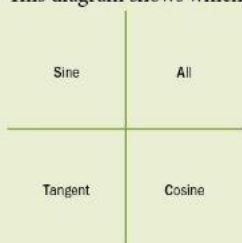
Exact values of trigonometric functions

Degrees	Radians	sin	cos	tan
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
45°	$\frac{\pi}{4}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

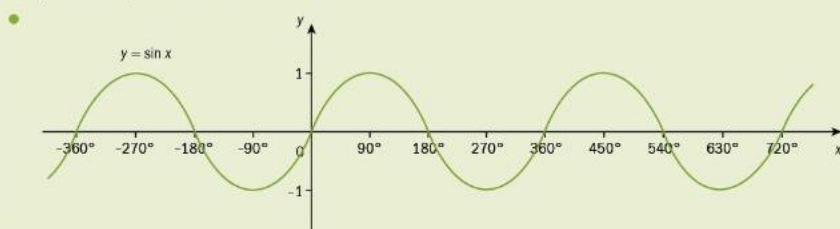
Angles in the range 0° to 360°

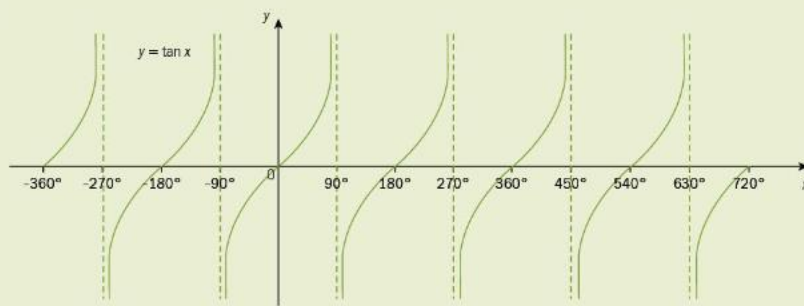
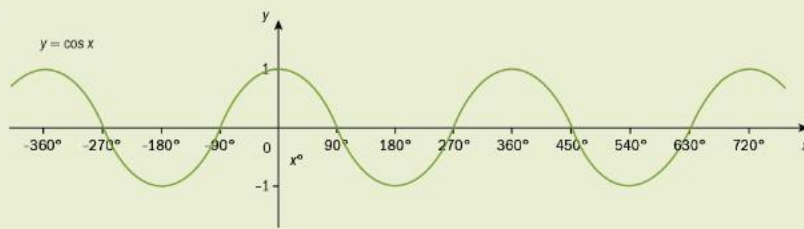
$\sin(180^\circ - \theta) = \sin \theta$
 $\cos(180^\circ - \theta) = -\cos \theta$
 $\tan(180^\circ - \theta) = -\tan \theta$
 $\sin(180^\circ + \theta) = -\sin \theta$
 $\cos(180^\circ + \theta) = -\cos \theta$
 $\tan(180^\circ + \theta) = \tan \theta$
 $\sin(360^\circ - \theta) = -\sin \theta$
 $\cos(360^\circ - \theta) = \cos \theta$
 $\tan(360^\circ - \theta) = -\tan \theta$

- This diagram shows which functions are positive in each quadrant.



Graphs of trigonometric functions





Composite graphs

- The graph of $y = \sin(x + c^\circ)$ is a translation of $y = \sin x$ by $\begin{pmatrix} -c \\ 0 \end{pmatrix}$.
- The graph of $y = \sin x + d$ is a translation of $y = \sin x$ by $\begin{pmatrix} 0 \\ d \end{pmatrix}$.
- The graph of $y = a \sin x$ is a stretch of $y = \sin x$, factor a , parallel to the y -axis.
- The graph of $y = \sin bx$ is a stretch of $y = \sin x$, factor $\frac{1}{b}$, parallel to the x -axis.

Trigonometric identities

- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $\sin^2 \theta + \cos^2 \theta = 1$

6) Binomial expansion
a) Binomial expansion

Example 3

Write down the 5th term, in ascending powers of x , in the expansion of $(1 + x)^7$.

5th term: $\binom{7}{4} 1^{7-4} x^4$

$= \frac{7!}{(7-4)!4!} x^4 = 35x^4$

5th term is the $(r + 1)$ th term so $r = 4$.

Use your calculator to get 35.

Example 4

Find the first three terms, in descending powers of x , in the expansion of $(4x - 3)^5$.

1st term $= 1(4x)^5 = 1024x^5$

2nd term $= \binom{5}{1} (4x)^4 (-3)^1$

$= 5(256x^4)(-3) = -3840x^4$

3rd term $= \binom{5}{2} (4x)^3 (-3)^2$

$= 10(64x^3)(9) = 5760x^3$

$(4x - 3)^5 = 1024x^5 - 3840x^4 + 5760x^3 + \dots$

The two indices add up to 5 for each term.

The two indices add up to 5 for each term.

Example 5

Find the coefficient of x^2 in the expansion of $(2 + 3x)^7$.

3rd term $= \binom{7}{2} (2)^5 (3x)^2$

$= 21(32)(9x^2) = 6048x^2$

Thus the coefficient is 6048.

1st term has no x , 2nd term has x , 3rd term has x^2 .

Remember to square 3 as well as x .

Do not put x^2 with the answer.

b) More complex expansion

Example 7

Find the first five terms, in descending powers of x , in the expansion of $(x-2)\left(x+\frac{3}{x}\right)^9$.

$$\left(x+\frac{3}{x}\right)^9 = (x)^9 + \binom{9}{1}(x)^8\left(\frac{3}{x}\right)^1 + \binom{9}{2}(x)^7\left(\frac{3}{x}\right)^2 + \dots \quad \leftarrow \text{First expand } \left(x+\frac{3}{x}\right)^9.$$

$$= x^9 + 27x^7 + 324x^5 + \dots$$

Note: We know that the highest power of x will be x^{10} from $x \times x^9$.
 The final expansion will have x^{10} to x^6 .
 x^6 comes from $x \times x^5$.
 Thus we only need to expand the 2nd bracket to the term in x^5 .

$$\begin{aligned} (x-2)\left(x+\frac{3}{x}\right)^9 &= x(x^9 + 27x^7 + 324x^5 + \dots) - 2(x^9 + 27x^7 + 324x^5 + \dots) \\ &= x^{10} + 27x^8 + 324x^6 + \dots - 2x^9 - 54x^7 - 648x^5 + \dots \\ &= x^{10} - 2x^9 + 27x^8 - 54x^7 + 324x^6 + \dots \end{aligned}$$

Example 8

Given that the coefficient of x^3 in the expansion of $(1+ax+2x^2)(2-x)^7$ is 560, determine the value of a .

$$\begin{aligned} (2-x)^7 &= 2^7 + \binom{7}{1}(2)^6(-x) + \binom{7}{2}(2)^5(-x)^2 + \binom{7}{3}(2)^4(-x)^3 + \dots \\ &= 128 - 448x + 672x^2 - 560x^3 + \dots \end{aligned}$$

$$\begin{aligned} \text{Terms in } x^3 \text{ in the expansion of } (1+ax+2x^2)(2-x)^7 \\ &= 1(-560x^3) + ax(672x^2) + 2x^2(-448x) \end{aligned}$$

$$\text{Coefficient of } x^3 = -560 + 672a - 896 = 672a - 1456 = 560$$

$$672a = 2016$$

$$a = 3$$

9. Find $(1+x)^4 - (1-x)^4$. Write your answer in ascending powers of x .
10. a) Find the first five terms in the expansion of $(2+x)^6$ in ascending powers of x .
b) Using your expansion, show that $(2.03)^6 = 69.9804$ correct to 4 decimal places.
11. Find the ratio of the coefficient of the x^3 term to the coefficient of the x^4 term in the expansion of $\left(2x + \frac{1}{2}\right)^6$.
12. Find the 5th term in the expansion of $\left(2 - \frac{1}{3x}\right)^{11}$ in ascending powers of x .
13. $(1+kx)^8 = 1 + 12x + ax^2 + bx^3 + \dots$
a) Find the value of k , the value of a and the value of b .
b) Use your values of k , a and b to find the coefficient of the term in x^3 in the expansion of $(1-x)(1+kx)^8$.
14. a) Expand $(1+2x)^3$.
b) Show that $(1+2x)^3 + (1-2x)^3 = 2(1+12x^2)$.
c) Hence solve $(1+2x)^3 - (1-2x)^3 = 8$.
15. a) Write down the first three terms, in ascending powers of x , in the expansion of $(1+ax)^n$ where $a \neq 0$ and $n > 2$.
b) The coefficient of the x^2 term is half the coefficient of the term in x . Show that $n = \frac{1+a}{a}$.
c) Find the coefficient of the x^2 term when $a = \frac{1}{5}$.
16. The first three terms, in ascending powers of x , in the expansion of $\left(1 + \frac{a}{x}\right)^n$ are $1 + \frac{24}{x} + \frac{252}{x^2}$. Find the value of a and the value of n .
17. a) Write down the binomial expansion of $(1+x)^5$.
b) Use your answer to (a) to express $(1-\sqrt{2})^5$ in the form $p + q\sqrt{2}$ where p and q are integers.
18. Find the coefficient of x^2 in the expansion of $(2-3x+x^2)\left(1-\frac{x}{2}\right)^{10}$.
19. $(2-ax)^8 = 2^b - 5120x + cx^2 + \dots$ Find the value of a , the value of b and the value of c .
20. $(1+x)^4(1+ax)^7 = 1 + 74x^2 + bx^3 + \dots$ Find the value of a and the value of b .

Summary:

Binomial expansion

- The **binomial expansion** of $(1+x)^n$ is $1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots + x^n$ where n is a positive integer.
- The coefficient of the term in x^r = the $(r+1)$ th term $= \binom{n}{r} = \frac{n!}{(n-r)!r!} = {}^nC_r$.
- $(x+y)^n = \binom{n}{0}x^ny^0 + \binom{n}{1}x^{n-1}y^1 + \binom{n}{2}x^{n-2}y^2 + \dots + \binom{n}{n-1}x^1y^{n-1} + \binom{n}{n}x^0y^n$

This is known as the **binomial theorem** where n is a positive integer.

7) Series

a) Arithmetic progressions

Example 5

The 1st term of an arithmetic progression is -6 and the common difference is 5 .

Find the 4th term and the n th term.

$$a = -6 \quad d = 5$$

$$4\text{th term} = a + 3d$$

$$= -6 + (3)5 = 9$$

Substitute $a = -6$ and $d = 5$

$$n\text{th term} = a + (n - 1)d$$

$$= -6 + (n - 1)5 = 5n - 11$$

Substitute $a = -6$ and $d = 5$

Example 6

The n th term of an arithmetic progression is $7n - 2$. Find the 1st term and the common difference, d .

The 1st term has $n = 1$

$$u_1 = 7(1) - 2 = 5$$

Substitute $n = 1$

The 2nd term has $n = 2$

$$u_2 = 7(2) - 2 = 12$$

Substitute $n = 2$

$d = 2\text{nd term} - 1\text{st term}$

$$d = 12 - 5 = 7$$

Example 7

The 5th term of an arithmetic progression is 21 and the sum of the first six terms is 90 .

Find the 18th term.

$$5\text{th term} = a + 4d = 21$$

$$S_6 = \frac{6}{2} [2a + (6 - 1)d] = 90$$

$$\text{Thus, } 6a + 15d = 90$$

Solving $a + 4d = 21$ and $6a + 15d = 90$ simultaneously gives $a = 5$ and $d = 4$.

$$18\text{th term} = a + 17d$$

$$= 5 + 17(4) = 73$$

Example 8

The sum of the first n terms of an arithmetic progression is $n^2 + 5n$.

a) Find the n th term. b) Write down the first four terms.

$$\text{a) } S_1 = 1^2 + 5(1) = 6 = \text{first term}$$

Substitute $n = 1$ into $n^2 + 5n$.

$$S_2 = 2^2 + 5(2) = 14 = \text{first term} + \text{second term}$$

$$\text{Second term} = S_2 - S_1 = 14 - 6 = 8$$

Substitute $n = 2$ into $n^2 + 5n$.

$$a = 6, d = 2,$$

$$n\text{th term} = 6 + (n - 1)2 = 2n + 4$$

Or

$$\text{The } n\text{th term} = S_n - S_{n-1} = n^2 + 5n - [(n - 1)^2 + 5(n - 1)]$$

For S_{n-1} put $n - 1$ instead of n .

$$= n^2 + 5n - [n^2 - 2n + 1 + 5n - 5]$$

$$= 2n + 4$$

$$\text{b) First four terms} = 2(1) + 4, 2(2) + 4, 2(3) + 4, 2(4) + 4$$

$$= 6, 8, 10, 12$$

Note: It is useful to remember that the n th term of any progression is $S_n - S_{n-1}$.

b) Geometric progression

Example 9

The 3rd term of a geometric progression is 9 and the common ratio is -3 .
Find the 6th term and the n th term.

$$\begin{array}{llll}
 \text{3rd term} & = ar^2 & = 9 & a \times (-3)^2 = 9 \quad a = \frac{9}{9} = 1 \quad \leftarrow \text{Substitute } r = -3. \\
 \text{6th term} & = ar^5 & = 1(-3)^5 & = -243 \\
 \text{\textit{n}th term} & = ar^{n-1} & = (1)(-3)^{n-1} & = (-3)^{n-1}
 \end{array}$$

Example 10

The n th term of a geometric progression is $10\left(\frac{1}{2}\right)^{n-1}$. Find the 1st term and the common ratio.

$$\begin{array}{ll}
 \text{1st term} = u_1 = 10\left(\frac{1}{2}\right)^0 = 10 & \leftarrow \text{Substitute } n = 1. \\
 a = 10 \\
 \text{2nd term} = u_2 = 10\left(\frac{1}{2}\right)^1 = 5 & \leftarrow \text{Substitute } n = 2. \\
 r = \text{2nd term} \div \text{1st term} = 5 \div 10 \\
 r = \frac{1}{2}
 \end{array}$$

The **sum of a geometric progression** can be written as $S_n = \frac{a(1-r^n)}{1-r}$, $r \neq 1$.

We use this formula when $-1 < r < 1$.

When we do not have $-1 < r < 1$, we use the formula $S_n = \frac{a(r^n-1)}{r-1}$.

Proof:

$$\begin{array}{ll}
 S_n = a + ar + ar^2 + \dots + ar^{n-3} + ar^{n-2} + ar^{n-1} \\
 rS_n = ar + ar^2 + \dots + ar^{n-3} + ar^{n-2} + ar^{n-1} + ar^n & \leftarrow \text{Multiply each term by } r. \\
 S_n - rS_n = a - ar^n & \leftarrow \text{Subtract the 2nd line from the 1st line.} \\
 S_n(1-r) = a(1-r^n) & \leftarrow \text{Factorise each side.} \\
 S_n = \frac{a(1-r^n)}{1-r} & \leftarrow \text{Use when } |r| < 1. \\
 \text{If we multiply by } \frac{-1}{-1} \text{ we get } S_n = \frac{a(r^n-1)}{r-1} & \leftarrow \text{Use when } |r| > 1.
 \end{array}$$

Note: This is not defined when $r = 1$.

Example 11

The 3rd term of a geometric progression is 16 and the 6th term is -128 .
Find the 1st term and the sum of the first seven terms.

$$\begin{array}{ll}
 ar^2 = 16 & ar^5 = -128 \\
 \frac{ar^5}{ar^2} = \frac{-128}{16} & \leftarrow \text{With geometric progressions we divide the equations to solve simultaneously.} \\
 r^3 = -8 & r = -2 \\
 ar^2 = 16 & a(-2)^2 = 16 \quad a = 16 \div 4 = 4 \\
 S_7 = \frac{a(r^7-1)}{r-1} = \frac{4((-2)^7-1)}{-2-1} & \leftarrow \text{Substitute } a = 4, r = -2 \text{ and } n = 6. \\
 = \frac{4(-128-1)}{-3} = 172
 \end{array}$$

Example 12

Mabintou deposits \$100 in a bank account at the start of each year.
She earns 4% compound interest.

- a) Work out how much money she has in her bank account at the end of the 3rd year.
- b) Work out how much money she has in her account at the start of the 30th year after she has made her annual deposit.

- a) At the end of the third year the amount in her account is
- $$\begin{aligned} & \$100(1.04) + \$100(1.04)^2 + \$100(1.04)^3 \\ & = \$104 + \$108.16 + \$112.4864 = \$324.65 \end{aligned}$$

- b) At the start of the 30th year, in \$, she has
- $$100 + 100(1.04) + 100(1.04)^2 + \dots + 100(1.04)^{29}$$
- $$a = 100 \quad r = 1.04 \quad n = 30$$

$$S_{30} = \frac{100(1.04^{30} - 1)}{1.04 - 1} = \$5608.49 \text{ (to the nearest cent)}$$

Example 13

An arithmetic progression has first term a and common difference d .

The 1st, 3rd and 13th terms of the arithmetic progression form the first three terms of a geometric progression with common ratio r .

- a) Express d in terms of a , and find the value of r .
- b) The 4th term in the geometric progression also appears in the arithmetic progression. Determine which term it is.

- a) $a, a + 2d, a + 12d$ are the first three terms in the geometric progression.

$$\text{So } (a + 2d)^2 = a(a + 12d)$$

$$a^2 + 4ad + 4d^2 = a^2 + 12ad$$

$$4d^2 = 8ad$$

$$d = 2a$$

$$\text{So } a + 2d = a + 2 \times 2a = 5a$$

The 2nd term of the geometric progression is $5a$, so $r = 5$.

- b) The 4th term in the geometric progression is $ar^3 = 125a$.

If this is the n th term of the arithmetic progression with first term a and common difference $2a$:

$$125a = a + (n - 1)(2a) \Rightarrow n = 63$$

So the 4th term in the geometric progression is the 63rd term in the arithmetic progression.

These are the 1st, 3rd and 13th terms of an arithmetic progression.

If a, b and c are consecutive terms in the geometric progression, then $\frac{b}{a} = \frac{c}{b}$, so $b^2 = ac$.

Expand the bracket and simplify the arithmetic progression.

Substitute $d = 2a$.

2nd term $\div$ 1st term $= 5a \div a$

Substitute $r = 5$.

c) Infinite geometric progressions

Example 14

The n th term of a geometric progression is $81 \left(-\frac{1}{3}\right)^n$. Find the sum to infinity.

$$\begin{aligned} \text{1st term: } u_1 &= 81 \left(-\frac{1}{3}\right)^1 = -27 \\ a &= -27 \end{aligned}$$

← Substitute $n = 1$.

$$\text{2nd term: } u_2 = 81 \left(-\frac{1}{3}\right)^2 = 9$$

← Substitute $n = 2$.

$$r = \frac{u_2}{u_1} = \frac{9}{-27} = -\frac{1}{3}$$

$$S_\infty = \frac{a}{1-r} = -\frac{27}{1 - \left(-\frac{1}{3}\right)} = -20\frac{1}{4}$$

← Substitute $a = -27$ and $r = -\frac{1}{3}$.

Example 15

a) Write the recurring decimal $0.4\dot{5}$ as the sum of a geometric progression.

b) Use your answer to part (a) to show that $0.4\dot{5}$ can be written as $\frac{41}{90}$.

$$\text{a) } 0.4\dot{5} = 0.4 + 0.0\dot{5} = 0.4 + \frac{5}{100} + \frac{5}{1000} + \frac{5}{10000} + \dots$$

$$\text{b) Consider } \frac{5}{100} + \frac{5}{1000} + \frac{5}{10000} + \dots \quad a = \frac{5}{100} \quad r = \frac{1}{10}$$

$$S_\infty = \frac{a}{1-r} = \frac{\frac{5}{100}}{1 - \frac{1}{10}} = \frac{5}{100} \times \frac{10}{9} = \frac{5}{90}$$

$$\text{Thus } 0.4\dot{5} = 0.4 + 0.0\dot{5} = \frac{2}{5} + \frac{5}{90} = \frac{41}{90}$$

Summary

Terms of a sequence

- u_n can be used to describe any term in a sequence. u_1 is the first term.
- The term after u_n is u_{n+1} .
- The term before u_n is u_{n-1} .

Finite and infinite series

- A series is **finite** if it has a finite number of terms, e.g. $5 + 9 + 13 + 17 + 21$.
- A series is **infinite** if it continues indefinitely, e.g. $5 + 9 + 13 + 17 + 21 + \dots$
- The **general term** of the series $5 + 9 + 13 + 17 + 21 + \dots$ is $4r + 1$.
- The finite series $5 + 9 + 13 + 17 + 21 + 25$ can be written as $\sum_{r=1}^6 (4r + 1)$.
- The infinite series $5 + 9 + 13 + 17 + 21 + \dots$ can be written as $\sum_{r=1}^{\infty} (4r + 1)$.

Arithmetic progressions

- a = 1st term ($= u_1$)
- d = common difference ($= u_2 - u_1 = u_3 - u_2 = u_4 - u_3$ etc.)
- l = last term in a finite series
- An arithmetic progression with n terms can be written as $a, (a + d), (a + 2d), (a + 3d), \dots, [a + (n - 1)d]$.
- The **n th term** of an arithmetic progression can be written as $[a + (n - 1)d]$.
- The **n th term** is $l = a + (n - 1)d$.
- We can use this formula to write down an expression for any term.
e.g. 8th term $= a + 7d$ e.g. 20th term $= a + 19d$.

Geometric progressions

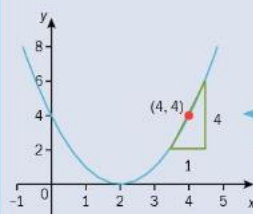
- A **geometric progression** is a series that has a **common ratio** between each term.
- a = 1st term ($= u_1$)
- r = common ratio ($= u_2 \div u_1 = u_3 \div u_2 = u_4 \div u_3$ etc.)
- l = last term in a finite series
- A geometric progression with n terms can be written as $a, ar, ar^2, ar^3, \dots, ar^{n-1}$.
- The **n th term** of a geometric progression can be written as ar^{n-1} .
- The **n th term** is $l = ar^{n-1}$.
- We can use this formula to write down an expression for any term.
e.g. 8th term $= ar^7$ e.g. 20th term $= ar^{19}$
- The **sum of a geometric progression** can be written as $S_n = \frac{a(1 - r^n)}{1 - r}$.
- The sum to infinity of a geometric progression is $S_{\infty} = \frac{a}{1 - r}$ provided $|r| < 1$.

8) Differentiation

a) Gradient of tangent

Example 1

Estimate the gradient of the tangent to the curve $y = (x - 2)^2$ at the point $(4, 4)$.



We can only *estimate* the gradient as it is difficult to draw the tangent accurately.

Hint: Draw the base of the triangle with an integer value.

Gradient of the tangent at $(4, 4) = \frac{4}{1} = 4$

b) Differentiation of polynomials

Example 3

Find the derived function in each case.

a) $f(x) = x^8$

b) $f(x) = \frac{6}{x^4}$

c) $f(x) = 3x$

d) $f(x) = 5$

Continued on the next page

a) $f(x) = x^8$

$$f'(x) = 8x^{8-1} = 8x^7$$

Use nx^{n-1} .

b) $f(x) = \frac{6}{x^4} = 6x^{-4}$

$$f'(x) = -24x^{-5}$$

$$= -\frac{24}{x^5}$$

First write the power as a negative power.

$-4 \times 6 = -24$ and -5 is one less than -4 .

c) $f(x) = 3x$

$$f'(x) = 3$$

$$f(x) = 3x^1, f'(x) = 3x^{1-1} = 3x^0 = 3 \times 1 = 3$$

d) $f(x) = 5 = 5x^0$

$$f'(x) = 0$$

Note: When we differentiate a constant, c , we always get 0. This makes sense as the graph of $f(x) = c$ is a horizontal straight line.

Example 4

Find $\frac{dy}{dx}$ when $y = 3\sqrt{x}$.

$$y = 3\sqrt{x} = 3x^{\frac{1}{2}}$$

First write $\sqrt{x}$ as $x^{\frac{1}{2}}$.

$$\frac{dy}{dx} = \frac{3}{2}x^{-\frac{1}{2}}$$

$$= \frac{3}{2\sqrt{x}}$$

$$x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$$

Example 5

Find $\frac{dy}{dx}$ when

a) $y = x^5 + 4x^4$

b) $y = (2x - 5)^2$

c) $y = \frac{3x^6 + 8}{x^2}$

a) $y = x^5 + 4x^4$

$$\frac{dy}{dx} = 5x^4 + 16x^3$$

Differentiate term by term.

b) $y = (2x - 5)^2 = 4x^2 - 20x + 25$

$$\frac{dy}{dx} = 8x - 20$$

First expand the brackets.

When you differentiate a constant it is 0.

c) $y = \frac{3x^6 + 8}{x^2} = \frac{3x^6}{x^2} + \frac{8}{x^2} = 3x^4 + 8x^{-2}$

$$\frac{dy}{dx} = 12x^3 - 16x^{-3}$$

$$= 12x^3 - \frac{16}{x^3}$$

First divide each term by x^2 .

c) Chain rule

Example 7

Find $\frac{dy}{du}$ when $y = \sqrt{4 - 3x}$.

Let $u = 4 - 3x$

$$\frac{du}{dx} = -3$$

$$y = \sqrt{u} = u^{\frac{1}{2}}$$

$$\frac{dy}{du} = \frac{1}{2} u^{-\frac{1}{2}}$$

$$= \frac{1}{2} (4 - 3x)^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{2} (4 - 3x)^{-\frac{1}{2}} \times -3$$

$$= -\frac{3}{2} (4 - 3x)^{-\frac{1}{2}}$$

$$= -\frac{3}{2\sqrt{4 - 3x}}$$

Let u = the expression in the square root.

Now substitute $u = 4 - 3x$.

Apply the chain rule.

Simplify your answer.

Example 8

Find $\frac{dy}{dx}$ when $y = \frac{2}{(3x^2 + 1)^4}$.

Let $u = 3x^2 + 1$

$$\frac{du}{dx} = 6x$$

$$y = 2u^{-4}$$

$$\frac{dy}{du} = -8u^{-5}$$

$$= -8(3x^2 + 1)^{-5}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = -8(3x^2 + 1)^{-5} \times 6x$$

$$= -48x(3x^2 + 1)^{-5}$$

$$= -\frac{48x}{(3x^2 + 1)^5}$$

Let u = the expression in the bracket.

Now substitute $u = 3x^2 + 1$.

Apply the chain rule.

Simplify your answer.

d) Gradient of tangent

Example 9

Find the gradient of the tangent to the curve $y = 4x^3 - 2x^2 + 1$ at the point $(1, 3)$.

$$\frac{dy}{dx} = 12x^2 - 4x$$

$$\text{At } x = 1, \frac{dy}{dx} = 12(1)^2 - 4(1) = 8$$

Gradient of the tangent = 8

$\frac{dy}{dx}$ gives us the gradient at any point on the curve.

We only need to use the x -value.

Example 10

Find the coordinates of the point on the curve with equation $y = 5 - 3x - 4x^2$ where the gradient is 1.

$$\frac{dy}{dx} = -3 - 8x = 1$$

$$-8x = 4$$

$$x = -0.5$$

When $x = -0.5$,

$$y = 5 - 3(-0.5) - 4(-0.5)^2 = 5.5$$

Coordinates are $(-0.5, 5.5)$.

Put $\frac{dy}{dx} = 1$ as the gradient = 1.

Substitute in the original equation to find y .

Example 11

Find the gradient of the tangent to the curve $y = 4x^2 - 5x + 2$ at each of the points where the curve meets the line $y = 7x - 6$.

The curve meets the line when

$$4x^2 - 5x + 2 = 7x - 6$$

$$4x^2 - 12x + 8 = 0$$

$$x^2 - 3x + 2 = 0$$

$$(x - 1)(x - 2) = 0$$

$$x = 1 \text{ or } x = 2$$

$$\frac{dy}{dx} = 8x - 5$$

When $x = 1$, the gradient of the tangent = $8(1) - 5 = 3$.

When $x = 2$, the gradient of the tangent = $8(2) - 5 = 11$.

The y -values are equal at the points of intersection.

It is easier to factorise if we divide the equation by the common factor of 4.

Differentiate to find an expression for the gradient.

e) Second derivative

Example 12

Find $\frac{d^2y}{dx^2}$ when $y = (4x^2 + 1)^2$.

$$y = (4x^2 + 1)(4x^2 + 1) = 16x^4 + 8x^2 + 1$$

$$\frac{dy}{dx} = 64x^3 + 16x$$

$$\frac{d^2y}{dx^2} = 192x^2 + 16$$

First expand the brackets.

Differentiate each of the three terms.

Then differentiate again.

Example 13

Find $\frac{d^2y}{dx^2}$ when $\frac{dy}{dx} = \frac{3}{\sqrt{x}}$.

$$\frac{dy}{dx} = \frac{3}{\sqrt{x}} = 3x^{-\frac{1}{2}}$$

$$\frac{d^2y}{dx^2} = -\frac{3}{2}x^{-\frac{3}{2}}$$

$$= -\frac{3}{2\sqrt{x^3}}$$

Remember $\frac{3}{\sqrt{x}} = \frac{3}{x^{\frac{1}{2}}} = 3x^{-\frac{1}{2}}$

We only need to differentiate once when given $\frac{dy}{dx}$.

$x^{\frac{3}{2}} = \sqrt{x^3}$ or $(\sqrt{x})^3$ or $x\sqrt{x}$

Example 14

Given that $f(x) = 8x^3 - 3x + \frac{4}{x}$, find the value of $f''(-2)$.

$$f'(x) = 24x^2 - 3 - \frac{4}{x^2}$$

$$f''(x) = 48x + \frac{8}{x^3}$$

$$\begin{aligned} f''(-2) &= 48(-2) + \frac{8}{(-2)^3} \\ &= -96 - 1 = -97 \end{aligned}$$

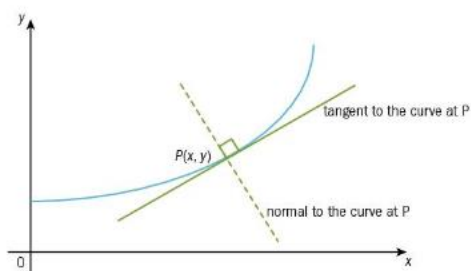
Differentiate $f(x)$ to get $f'(x)$.

Differentiate $f'(x)$ to get $f''(x)$.

Substitute $x = -2$ in to $f''(x)$.

f) Equation of tangents and normal

We can use differentiation to find the gradient of the tangent to a curve at a particular point. We will then be able to find the **equation of the tangent** and the **equation of the normal** at this point by using our knowledge of coordinate geometry.



Example 15

Find the equation of the tangent to the curve $y = 3x^3 - 6x^2 + x$ at the point $(1, -2)$.

$$y = 3x^3 - 6x^2 + x$$

$$\frac{dy}{dx} = 9x^2 - 12x + 1$$

$$\text{When } x = 1, \frac{dy}{dx} = 9(1)^2 - 12(1) + 1 = -2.$$

Tangent goes through $(1, -2)$ with $m = -2$.

$$y + 2 = -2(x - 1)$$

$$y + 2 = -2x + 2$$

Equation of tangent is $y + 2x = 0$.

Differentiate to find the gradient of the tangent.

Substitute to find the gradient at $(1, -2)$.

We use $(1, -2)$ and $m = -2$ to find the equation.

Example 16

Find the equation of the normal to the curve $y = 4\sqrt{x} + 7$ at the point where $x = 16$.

$$y = 4\sqrt{x} + 7 = 4x^{\frac{1}{2}} + 7$$

Write the expression using indices.

$$\frac{dy}{dx} = 2x^{-\frac{1}{2}} = \frac{2}{\sqrt{x}}$$

$$\text{When } x = 16, y = 4\sqrt{16} + 7 = 23$$

Substitute $x = 16$ into $y = 4\sqrt{x} + 7$.

$$\text{and } \frac{dy}{dx} = \frac{2}{\sqrt{16}} = \frac{1}{2} \text{ so gradient of normal} = -2.$$

Gradient of normal = $-\frac{1}{\text{gradient of tangent}}$

Normal goes through $(16, 23)$ with $m = -2$.

$$y - 23 = -2(x - 16)$$

$$\text{Equation of normal is } y = -2x + 55.$$

Example 17

The equation of a curve is $y = 4x - x^2$. The normal to the curve at the point $P(1, 3)$ meets the curve again at Q . Find the coordinates of Q .

$$y = 4x - x^2$$

$$\frac{dy}{dx} = 4 - 2x$$

$$\text{When } x = 1, \frac{dy}{dx} = 4 - 2 = 2,$$

$$\text{so gradient of the normal} = -\frac{1}{2}.$$

$$\text{Equation of the normal is } y - 3 = -\frac{1}{2}(x - 1).$$

$$\text{So } y = -\frac{1}{2}x + \frac{7}{2}$$

To find where the curve meets the normal

we solve $y = 4x - x^2$ and $y = -\frac{1}{2}x + \frac{7}{2}$ simultaneously.

$$4x - x^2 = -\frac{1}{2}x + \frac{7}{2}$$

$$8x - 2x^2 = -x + 7$$

Multiply both sides by 2.

$$2x^2 - 9x + 7 = 0$$

$$(2x - 7)(x - 1) = 0$$

$$x = \frac{7}{2}, y = \frac{7}{4} \text{ or } x = 1, y = 3$$

We already know it goes through $(1, 3)$.

The normal meets the curve again at $\left(\frac{7}{2}, \frac{7}{4}\right)$.

g) Stationary points

Example 3

Find the coordinates of the stationary points of the curve $y = 6x - 2x^3$ and **determine** their nature. **Sketch** the curve.

$$\frac{dy}{dx} = 6 - 6x^2 = 0 \text{ at stationary points}$$

Differentiate and put $\frac{dy}{dx}$ equal to 0.

$$6 = 6x^2$$

Solve for x .

$$x = 1 \text{ and } x = -1$$

$$y = 4 \text{ and } y = -4$$

Substitute x in to $y = 6x - 2x^3$ to find y .

x	-2	-1	0	1	2
$\frac{dy}{dx}$	-18	0	6	0	-18

Use values for x less than -1, between 0 and 1, and greater than 1.



Draw the positive and negative gradients.

(1, 4) is a maximum turning point.

Write down if it is a maximum or minimum.

(-1, -4) is a minimum turning point.

Note: $\frac{d^2y}{dx^2} = -12x$

This is an alternative method. It is useful when $\frac{dy}{dx}$ is simple to differentiate again.

When $x = 1$, $\frac{d^2y}{dx^2} = -12$, $\frac{d^2y}{dx^2} < 0$, so maximum.

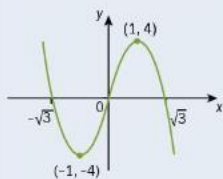
When $x = -1$, $\frac{d^2y}{dx^2} = 12$, $\frac{d^2y}{dx^2} > 0$, so minimum.

Sketch:

$$y = 6x - 2x^3 = 2x(3 - x^2)$$

Cuts x -axis at $0, \sqrt{3}, -\sqrt{3}$

Find when $y = 0$ for points where the graph crosses the x -axis.



Example 1

Find the values of x for which $f(x) = 5 - 4x - x^2$ is a decreasing function.

$$f'(x) = -4 - 2x$$

First differentiate to give $f'(x)$.

Decreasing function so $-4 - 2x < 0$.

$$-4 < 2x$$

Then solve for x in an inequality.

$$x > -2$$

$f(x)$ is a decreasing function when $x > -2$.

Example 2

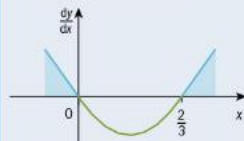
$y = x^3 - x^2$. Find the range of values of x for which y is increasing.

$$\frac{dy}{dx} = 3x^2 - 2x$$

First differentiate to find $\frac{dy}{dx}$.

$$= x(3x - 2) > 0$$

y is increasing so put $\frac{dy}{dx} > 0$.



Draw a diagram. Note that this is the graph of $\frac{dy}{dx}$.

$$x < 0 \text{ or } x > \frac{2}{3}$$

$\frac{dy}{dx} > 0$ so find where the graph lies above the x -axis, giving two intervals.

y is increasing when $x < 0$ and $x > \frac{2}{3}$.

Example 4

Find the coordinates of the stationary points of the curve $y = \frac{2}{2x-1} + x$ and determine their nature.

Let $A = \frac{2}{2x-1}$ and let $u = 2x - 1$

Use the chain rule to differentiate $\frac{2}{2x-1}$.

$$\frac{du}{dx} = 2$$

Let $A = \frac{2}{u} = 2u^{-1}$ $\frac{dA}{du} = -2u^{-2} = \frac{-2}{(2x-1)^2}$

$$\frac{dA}{dx} = \frac{-2}{(2x-1)^2} (2) = \frac{-4}{(2x-1)^2}$$

$$\frac{dy}{dx} = \frac{-4}{(2x-1)^2} + 1 = 0 \text{ at stationary points}$$

Put $\frac{dy}{dx} = 0$.

$$(2x-1)^2 = 4$$

Multiply each term by $(2x-1)^2$ and rearrange.

$$4x^2 - 4x - 3 = 0$$

$$(2x+1)(2x-3) = 0$$

$$x = -\frac{1}{2}, y = -\frac{3}{2} \quad \text{or} \quad x = \frac{3}{2}, y = \frac{5}{2}$$

x	-1	$-\frac{1}{2}$	0	$\frac{3}{2}$	2
$\frac{dy}{dx}$	$\frac{5}{9}$	0	-3	0	$\frac{5}{9}$

Draw up a table to show the gradient of the tangent (or we could find $\frac{d^2y}{dx^2}$ etc.).

$\left(-\frac{1}{2}, -\frac{3}{2}\right)$ is a maximum turning point.

Draw the positive and negative gradients and determine the nature of the stationary points.

$\left(\frac{3}{2}, \frac{5}{2}\right)$ is a minimum turning point.

h) Rates of changes

Example 7

The radius, r , of a circle is increasing at the rate of $\frac{2}{r^2} \text{ m s}^{-1}$.

Find the rate at which the area, A , is increasing when $r = 8$.

Follow the steps on the right of the examples for problems involving connected rates of change.

We want to find $\frac{dA}{dt}$.

Write down the rate of change we want to find.

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

Use the chain rule to write down an identity for $\frac{dA}{dt}$.

$$\frac{dr}{dt} = \frac{2}{r^2}$$

Write the information given as the rate of change of r with respect to t .

But, $A = \pi r^2$

Write down the link between A and r .

$$\frac{dA}{dr} = 2\pi r$$

Differentiate to the rate of change of A with respect to r .

$$\text{So, } \frac{dA}{dt} = 2\pi r \times \frac{2}{r^2} = \frac{4\pi}{r}$$

Substitute $\frac{dA}{dr}$ and $\frac{dr}{dt}$.

$$\text{When } r = 8, \frac{dA}{dt} = \frac{4\pi}{8}$$

We can now substitute $r = 8$.

$$\frac{dA}{dt} = \frac{\pi}{2}$$

$$= 1.57 \text{ m}^2 \text{ s}^{-1} \text{ (3 s.f.)}$$

Do not forget the units (area/time).

Example 8

A spherical balloon is being blown up so that its volume increases at a rate of $3 \text{ cm}^3 \text{ s}^{-1}$.

Find the rate of increase of the radius of the balloon when the volume of the balloon is 60 cm^3 .

We want to find $\frac{dr}{dt}$.

Write down the rate of change we want to find.

$$\frac{dr}{dt} = \frac{dr}{dV} \times \frac{dV}{dt}$$

Write down the chain rule for $\frac{dr}{dt}$.

$$\frac{dV}{dt} = 3$$

Write the information given as a rate of change.

$$V = \frac{4}{3}\pi r^3, \frac{dV}{dr} = 4\pi r^2$$

Differentiate to get the rate of change of V with respect to r .

$$\frac{dr}{dt} = \frac{1}{4\pi r^2} \times 3 = \frac{3}{4\pi r^2}$$

$$\frac{dr}{dV} = 1 \div \frac{dV}{dr}$$

$$\text{When } V = 60, 60 = \frac{4}{3}\pi r^3$$

Substitute for these rates of change.

$$r^3 = \frac{45}{\pi}$$

$$r = 2.42859\dots$$

We can now substitute for $r = 2.42859\dots$

$$\frac{dr}{dt} = \frac{3}{4\pi r^2} = \frac{3}{4\pi(2.42859\dots)^2}$$

$$= 0.0405 \text{ cm s}^{-1} \text{ (3 s.f.)}$$

Do not forget the units (radius/time).

Summary:

Differentiating $y = x^n$

- If $y = x^n$ then $\frac{dy}{dx} = nx^{n-1}$

Differentiating $y = ax^n$

- If $y = ax^n$ then $\frac{dy}{dx} = anx^{n-1}$

We differentiate when

- we are given $f(x)$ and want to find $f'(x)$
- we are given y and are want to find $\frac{dy}{dx}$
- we want to find $\frac{d}{dx}(f(x))$
- we want to find the gradient of the tangent (or the normal) of a curve given the equation of the curve
- we want to find the coordinates of a point on a curve and are given the value of the gradient at that point
- we want to find the rate of change of y with respect to x given y .

The chain rule (differentiating a function of a function)

- The chain rule states that:
If y is a function of u , and u is a function of x , $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

The second derivative

- If we differentiate $\frac{dy}{dx}$ we get $\frac{d^2y}{dx^2}$.
- If we differentiate $f'(x)$ we get $f''(x)$.

Increasing and decreasing functions

- When the gradient of a function is positive we say the function is an **increasing function**.
A function $f(x)$ is increasing for $a < x < b$ if $f'(x) > 0$ for $a < x < b$.
- When the gradient of a function is negative we say the function is a **decreasing function**.
A function $f(x)$ is decreasing for $a < x < b$ if $f'(x) < 0$ for $a < x < b$.

Stationary points

- Any point on a curve where $\frac{dy}{dx} = 0$ is called a **stationary point**.
- There are three types of stationary point: minimum points, maximum points and points of inflexion.
- At a minimum point, $\frac{d^2y}{dx^2} > 0$.
- At a maximum point, $\frac{d^2y}{dx^2} < 0$.
- At $\frac{d^2y}{dx^2} = 0$, the point may be a minimum, or a maximum, or a point of inflexion, and so we must look at the sign of $\frac{dy}{dx}$ either side of the stationary point to determine its nature.

Points of inflexion are not included in Cambridge International AS & A Level 9709 examination papers.

Problems involving maximum and minimum values

- We can use these steps to solve problems:
 - Draw a diagram where possible.
 - Choose letters to represent the unknown quantities.
 - Write an expression for the quantity we are trying to maximise/minimise – it may contain two unknowns.
 - Use a given condition from the question to write an equation connecting two variables.
 - Rearrange the equation to express one variable in terms of the other.
 - Use substitution to find an expression in terms of one variable only.
 - Use differentiation to find maximum/minimum values.

Connected rates of change

- Always derive a formula related to the rates of change given
 - e.g. $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ e.g. $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$
 - e.g. $\frac{dV}{dt} = \frac{dV}{ds} \times \frac{ds}{dt}$ e.g. $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dt} \times \frac{dt}{dx}$
- You can use these steps to solve problems:
 - Write the information given as a rate of change.
 - Write down the rate of change you want to find.
 - Write down the link between the two variables.
 - Differentiate to get another rate of change.
 - You may also need to use another formula and then differentiate this to get another rate of change.
 - Substitute these rates of change.
 - You can then substitute any given values to get the answer.

9) Integration

a) Integration

$$\text{If } \frac{dy}{dx} = x^n, \text{ then } y = \int x^n dx = \frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$$

$$\text{If } \frac{dy}{dx} = ax^n, \text{ then } y = \int ax^n dx = \frac{ax^{n+1}}{n+1} + c \quad (n \neq -1)$$

Example 1

Given $\frac{dy}{dx} = 9x^2 - 4x + 3x^{\frac{1}{2}} - 7$, find y .

$$y = \int (9x^2 - 4x + 3x^{\frac{1}{2}} - 7) dx \quad \leftarrow \text{We need to integrate } \frac{dy}{dx} \text{ to get } y.$$

$$= \frac{9x^3}{3} - \frac{4x^2}{2} + \frac{3x^{\frac{3}{2}}}{\frac{3}{2}} - 7x + c \quad \leftarrow \text{Use } \int x^n dx = \frac{x^{n+1}}{n+1} \text{ for each term.}$$

$$= 3x^3 - 2x^2 + 2x^{\frac{3}{2}} - 7x + c \quad \leftarrow \text{Simplify where possible and include '+ c' in your answer.}$$

b) Finding constant of integration

Example 3

Find the equation of the curve that passes through the point $(4, -1)$ and where

$$\frac{dy}{dx} = \frac{5x^3 + 1}{\sqrt{x}}$$

$$\frac{dy}{dx} = \frac{5x^3 + 1}{\sqrt{x}} = 5x^{\frac{5}{2}} + x^{-\frac{1}{2}} \quad \leftarrow \text{Divide each term by } x^{\frac{1}{2}}.$$

$$y = \frac{5x^{\frac{7}{2}}}{\frac{7}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + c \quad \leftarrow \text{Integrate each term and include '+ c' at the end.}$$

$$= 2x^{\frac{7}{2}} + 2x^{\frac{1}{2}} + c \quad \leftarrow \text{Dividing by } \frac{5}{2} \text{ is the same as multiplying by } \frac{2}{5}.$$

As $(4, -1)$ lies on the curve,

$$-1 = 2(4)^{\frac{7}{2}} + 2(4)^{\frac{1}{2}} + c \quad \leftarrow \text{Substitute } x = 4, y = -1.$$

$$c = -1 - 64 - 4 = -69 \quad \leftarrow \text{Find the value of } c.$$

$$y = 2x^{\frac{7}{2}} + 2x^{\frac{1}{2}} - 69 \quad \leftarrow \text{Write down the equation of the curve.}$$

Note: You can check you have integrated correctly by differentiating,

$$\text{e.g. } \frac{d}{dx} \left(2x^{\frac{7}{2}} + 2x^{\frac{1}{2}} + c \right) = 5x^{\frac{5}{2}} + x^{-\frac{1}{2}}$$

Example 4

A curve is such that $f'(x) = \frac{10}{x^3} - 4$ and the point $(-1, 2)$ lies on the curve.
Find the equation of the curve.

$f(x) = \int \left(\frac{10}{x^3} - 4 \right) dx = \int (10x^{-3} - 4) dx$	←	Integrate $f'(x)$ to find $f(x)$ by expressing each term as a power of x .
$= \frac{10x^{-2}}{-2} - 4x + c$	←	Integrate term by term.
$= -\frac{5}{x^2} - 4x + c$	←	Simplify and write your negative powers as fractions.
$2 = -\frac{5}{(-1)^2} - 4(-1) + c$	←	Substitute for $x = -1$ and $y = 2$.
$c = 3$	←	Find the value of c .
$f(x) = -\frac{5}{x^2} - 4x + 3$	←	Write down the equation of the curve.

Example 5

A curve is such that $\frac{d^2y}{dx^2} = -6x$ and the curve has a maximum point at $(1, 2)$.
Find the equation of the curve.

$\frac{dy}{dx} = \int -6x dx$	←	Integrate $\frac{d^2y}{dx^2}$ to find $\frac{dy}{dx}$.
$= -3x^2 + c$		
$0 = -3(1)^2 + c$	←	When $x = 1$ and $y = 2$ we have a maximum point, and so $\frac{dy}{dx} = 0$.
$c = 3$		
$\frac{dy}{dx} = -3x^2 + 3$	←	We now have an equation for $\frac{dy}{dx}$.
$y = -x^3 + 3x + k$	←	
$2 = -(1)^3 + 3(1) + k$	←	Integrate to find y .
$k = 0$	←	$y = 2$ when $x = 1$.
$y = -x^3 + 3x$	←	We can now write an equation for the curve.

Example 6Find $\int (4x - 3)^5 dx$.

$$\begin{aligned}\int (4x - 3)^5 dx &= \frac{1}{4} \frac{(4x - 3)^{5+1}}{(5+1)} + c \\ &= \frac{(4x - 3)^6}{24} + c\end{aligned}$$

Use the standard result with $a = 4$, $b = -3$, $n = 5$.**Example 7**Find $\int \frac{4}{(1-2x)^7} dx$.

$$\begin{aligned}\int \frac{4}{(1-2x)^7} dx &= 4 \int (1-2x)^{-7} dx \\ &= 4 \times \frac{1}{-2} \times \frac{(1-2x)^{-6}}{-6} + c \\ &= \frac{(1-2x)^{-6}}{3} + c \\ &= \frac{1}{3(1-2x)^6} + c\end{aligned}$$

We can take the constant factor 4 outside the integral.

 $a = 1$, $b = -2$, $n = -7$ **Example 8**Find $\int \frac{6}{\sqrt{4x+9}} dx$.

$$\begin{aligned}\int \frac{6}{\sqrt{4x+9}} dx &= \int 6(4x+9)^{-\frac{1}{2}} dx \\ &= 6 \times \frac{1}{4} \times \frac{(4x+9)^{\frac{1}{2}}}{\frac{1}{2}} + c \\ &= 3\sqrt{4x+9} + c\end{aligned}$$

Write in a form using negative powers.

Use the standard result with $a = 4$, $b = 9$, $n = -\frac{1}{2}$.

Simplify your answer.

c) Definite integrals

Example 9

- a) Find $\int (3x - 2)^8 dx$. b) Hence find $\int_0^1 (3x - 2)^8 dx$.

$$\begin{aligned} \text{a) } \int (3x - 2)^8 dx &= \frac{1}{3} \frac{(3x - 2)^9}{9} + c \\ &= \frac{(3x - 2)^9}{27} + c \end{aligned}$$

Also divide by the coefficient of x .

$$\begin{aligned} \text{b) } \int_0^1 (3x - 2)^8 dx &= \left[\frac{(3x - 2)^9}{27} \right]_0^1 \\ &= \frac{(3(1) - 2)^9}{27} - \frac{(3(0) - 2)^9}{27} \\ &= \frac{1}{27} + \frac{512}{27} = \frac{513}{27} = 19 \end{aligned}$$

We use square brackets to show we have integrated and we write the limits in the positions shown.

Calculate $f(b) - f(a)$.

Example 10

Find $\int_1^4 \frac{x^2 + x^3}{\sqrt{x}} dx$.

$$\begin{aligned} \int_1^4 \frac{x^2 + x^3}{\sqrt{x}} dx &= \int_1^4 \left(x^{\frac{3}{2}} + x^{\frac{5}{2}} \right) dx \\ &= \left[\frac{2}{5} x^{\frac{5}{2}} + \frac{2}{7} x^{\frac{7}{2}} \right]_1^4 \\ &= \left(\frac{2}{5} 4^{\frac{5}{2}} + \frac{2}{7} 4^{\frac{7}{2}} \right) - \left(\frac{2}{5} 1^{\frac{5}{2}} + \frac{2}{7} 1^{\frac{7}{2}} \right) \\ &= \left(\frac{2}{5} \times 32 + \frac{2}{7} \times 128 \right) - \left(\frac{2}{5} + \frac{2}{7} \right) \\ &= 12 \frac{2}{5} + 36 \frac{2}{7} = 48 \frac{24}{35} \end{aligned}$$

Divide each term in the numerator by $x^{\frac{1}{2}}$.

Substitute $x = 4$ and $x = 1$.

Evaluate as separate brackets.

Exercise 10.4

1. Find

a) $\int_0^1 6x dx$

b) $\int_2^5 x^2 dx$

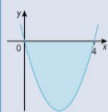
c) $\int_4^9 \sqrt{x} dx$

d) $\int_{-1}^1 \frac{x}{2} dx$

e) $\int_1^3 \frac{3}{x^2} dx$

f) $\int_1^4 \frac{1}{\sqrt{x}} dx$

d) Area bounded between two curves

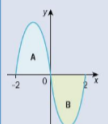
Example 12Find the area bounded by the curve with equation $y = x(x - 4)$ and the x -axis.

First draw a sketch.

$$\begin{aligned}\text{Area} &= \int_0^4 (x^2 - 4x) \, dx \\ &= \left[\frac{1}{3}x^3 - 2x^2 \right]_0^4 \\ &= \left(\frac{64}{3} - 32 \right) - (0 - 0) = -10\frac{2}{3} \\ \text{Area} &= 10\frac{2}{3}\end{aligned}$$

Note: Where the value of the integral is negative this shows that the region is below the x -axis.

Give the area as a positive value.

Example 13Find the area bounded by the curve with equation $y = x(x + 2)(x - 2)$ and the x -axis.

First draw a sketch.

We find the area by considering the two separate regions A and B.

$$\begin{aligned}\text{Area A} &= \int_{-2}^0 (x^3 - 4x) \, dx \\ &= \left[\frac{1}{4}x^4 - 2x^2 \right]_{-2}^0 \\ &= (0 - 0) - (4 - 8) = 4\end{aligned}$$

Expand the brackets.

► Continued on the next page

$$\begin{aligned}\text{Area B} &= \int_0^4 (x^3 - 4x) \, dx \\ &= \left[\frac{1}{4}x^4 - 2x^2 \right]_0^4 \\ &= (4 - 8) - (0 - 0) = -4\end{aligned}$$

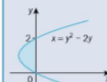
We can use the result from region A above.

The negative value shows the area is under the x -axis.

$$\text{Total area} = 4 + 4 = 8$$

Note that if we calculate $\int_{-2}^4 (x^3 - 4x) \, dx$ we get 0 as the answer.We can also use this technique to find the area bounded by a curve, $x = f(y)$, the y -axis and the lines $y = a$ and $y = b$.The area between the curve $x = f(y)$, the y -axis, and the lines $y = a$ and $y = b$ is given by:

$$\text{Area} = \int_a^b x \, dy \quad \text{or} \quad \int_a^b f(y) \, dy$$

Note: We usually write $\int_a^b$ instead of $\int_{y=a}^{y=b}$ **Example 14**Find the area bounded by the curve with equation $x = y^2 - 2y$ and the y -axis as shown in the diagram.

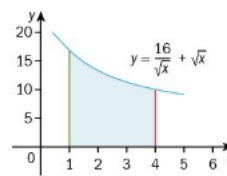
$$\begin{aligned}\text{Area} &= \int_0^2 x \, dy \\ &= \int_0^2 (y^2 - 2y) \, dy \\ &= \left[\frac{1}{3}y^3 - y^2 \right]_0^2 \\ &= \left(\frac{8}{3} - 4 \right) - (0 - 0) = -\frac{4}{3}\end{aligned}$$

Use the correct formula.

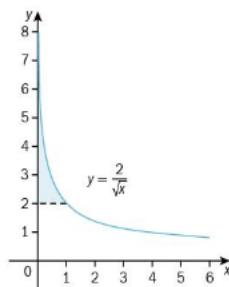
Everything is using y , including the limits.The negative value shows that the required area is to the left of the y -axis.We write that the required area is $\frac{4}{3}$.

Write as a positive value.

8. The diagram shows part of the curve with equation $y = \frac{16}{\sqrt{x}} + \sqrt{x}$.
Find the area bounded by the curve, the lines $x = 1$, $x = 4$ and the x -axis.

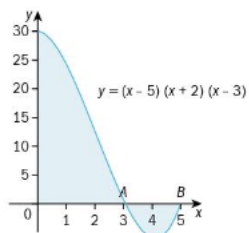


9.



Find the area of the shaded region.

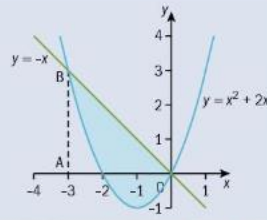
10. The diagram shows part of the curve with equation $y = (x - 5)(x + 2)(x - 3)$.
- Write down the coordinates of A and B .
 - Show that the equation may be written as $y = x^3 - 6x^2 - x + 30$.
 - Find the total shaded area.



Example 16

The diagram shows the curve $y = x^2 + 2x$ and the line $y = -x$.

Work out the area of the shaded region.



The curve meets the line when $x^2 + 2x = -x$

$$x^2 + 3x = 0$$

$$x(x + 3) = 0$$

Solve the two equations simultaneously to find where the line intersects the curve.

They meet at $(0, 0)$ and $(-3, 3)$

$$\text{Area of triangle OAB} = \frac{1}{2} (3 \times 3) = \frac{9}{2}$$

Area under the curve, between $x = -3$ and $x = -2$

$$= \int_{-3}^{-2} (x^2 + 2x) dx$$

The dotted line is at $x = -3$.

$$= \left[\frac{1}{3}x^3 + x^2 \right]_{-3}^{-2}$$

$$= \left(-\frac{8}{3} + 4 \right) - (-9 + 9)$$

$$= \frac{4}{3}$$

$$\text{Area of shaded region above the } x\text{-axis} = \frac{9}{2} - \frac{4}{3} = \frac{19}{6}$$

Area of shaded region below the x -axis

$$= \int_{-2}^0 (x^2 + 2x) dx$$

This is the area between the curve and the x -axis.

$$= \left[\frac{1}{3}x^3 + x^2 \right]_{-2}^0$$

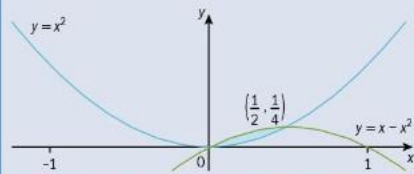
$$= (0 + 0) - \left(-\frac{8}{3} + 4 \right) = -\frac{4}{3}$$

The area is a negative value as it is below the curve.

$$\text{Area required} = \frac{19}{6} + \frac{4}{3} = \frac{9}{2}$$

Example 17

Find the area of the finite region bounded by $y = x^2$ and $y = x - x^2$.



$$\text{Area} = \int_0^{\frac{1}{2}} (x - x^2) \, dx - \int_0^{\frac{1}{2}} x^2 \, dx$$

Area under $y = x - x^2$ minus area under $y = x^2$

$$= \int_0^{\frac{1}{2}} (x - x^2 - x^2) \, dx$$

Combine the two integrals.

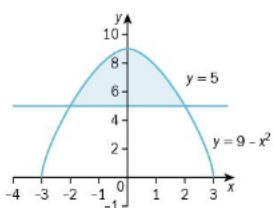
$$= \int_0^{\frac{1}{2}} (x - 2x^2) \, dx$$

$$= \left[\frac{1}{2}x^2 - \frac{2}{3}x^3 \right]_0^{\frac{1}{2}}$$

$$= \left(\frac{1}{8} - \frac{1}{12} \right) - (0 - 0)$$

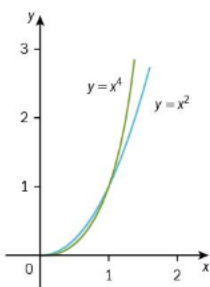
$$= \frac{1}{24}$$

3.



The diagram shows the curve $y = 9 - x^2$ and the line $y = 5$.
Find the shaded area.

4.



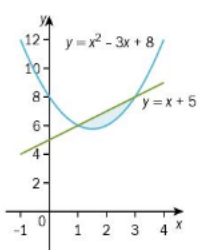
The diagram shows parts of the curves $y = x^2$ and $y = x^4$.
Find the shaded area.

5. Find the area between the curve with equation $y = x(x - 1)$ and the line $y = 3x$.

Find the coordinates of the points of intersection and sketch the graphs.

6. a) Show that the curve $y = 2x^2 + 3$ and the curve $y = 10x - x^2$ meet at the points where $x = 3$ and $x = \frac{1}{3}$.
b) Find the area between these two curves.

7.



Find the area bounded by the curve $y = x^2 - 3x + 8$ and the line $y = x + 5$.

e) Improper integrals

For (1) i)

We define the **improper integral** $\int_a^{\infty} f(x) dx$ as $\lim_{b \rightarrow \infty} \int_a^b f(x) dx$, provided the limit exists.

ii)

We define the **improper integral** $\int_{-\infty}^b f(x) dx$ as $\lim_{a \rightarrow -\infty} \int_a^b f(x) dx$, provided the limit exists.

For (2)

When $f(x)$ is defined for $0 < x < b$, but $f(x)$ is not defined when $x = 0$, then the **improper integral** $\int_0^b f(x) dx = \lim_{a \rightarrow 0^+} \int_a^b f(x) dx$, provided the limit exists.

Note: $\lim_{a \rightarrow 0^+}$ denotes 'a tends to 0 from just above 0'.

Example 18

Show that the improper integral $\int_1^{\infty} \frac{1}{x^2 \sqrt{x}} dx$ has a value and find that value.

$$\int_1^b \frac{1}{x^2 \sqrt{x}} dx = \int_1^b x^{-\frac{5}{2}} dx$$

$$= \left[-\frac{2}{3x^{\frac{3}{2}}} \right]_1^b$$

$$= \left(-\frac{2}{3b^{\frac{3}{2}}} \right) - \left(-\frac{2}{3} \right)$$

$$\text{As } b \rightarrow \infty, \frac{2}{3b^{\frac{3}{2}}} \rightarrow 0$$

$$\text{Thus } \int_1^{\infty} \frac{1}{x^2 \sqrt{x}} dx = -0 + \frac{2}{3}$$

Hence the improper integral does exist (as it has a finite value) and has a value of $\frac{2}{3}$.

Write the integral with an upper limit, say b (where b is a large number).

Integrate the function in the normal way.

As b tends to ∞ , the denominator gets larger and the value of the fraction tends to 0.

Example 19

Show that only one of the following improper integrals has a finite value and find that value.

a) $\int_{-2}^0 \frac{2}{x^5} dx$

b) $\int_{-\infty}^{-2} \frac{2}{x^5} dx$

$$\text{a) } \int_{-2}^p \frac{2}{x^5} dx = \left[-\frac{2}{4x^4} \right]_{-2}^p$$

$$= \left(-\frac{1}{2p^4} \right) - \left(-\frac{1}{2(-2)^4} \right)$$

$$\text{As } p \rightarrow 0, \frac{1}{2p^4} \text{ has no finite value}$$

$$\text{and so, } -\frac{1}{2p^4} + \frac{1}{32} \text{ has no finite value.}$$

$$\text{b) } \int_q^{-2} \frac{2}{x^5} dx = \left[-\frac{2}{4x^4} \right]_q^{-2}$$

$$= \left(-\frac{1}{2(-2)^4} \right) - \left(-\frac{1}{2q^4} \right)$$

$$\text{As } q \rightarrow -\infty, \frac{1}{2q^4} \rightarrow 0$$

$$\text{Thus } \int_{-\infty}^{-2} \frac{2}{x^5} dx = -\frac{1}{32} + 0$$

Hence, this improper integral does exist and has a value of $-\frac{1}{32}$.

As the function to be integrated is not defined at $x = 0$, replace the upper limit 0 with a letter, say p ($-2 < p < 0$ and p just less than 0).

As $p \rightarrow 0$, $\frac{1}{2p^4} \rightarrow \infty$ and so has no finite value.

We can say the integral does not exist.

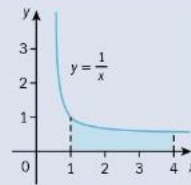
Start by replacing $-\infty$ with a variable, say q .

3. a) Find, in terms of a and b , the value of the integral $\int_a^b \frac{6}{x^{\frac{5}{2}}} dx$.
- b) Show that only one of the following improper integrals has a finite value and find that value.
- i) $\int_0^9 \frac{6}{x^{\frac{5}{2}}} dx$ ii) $\int_9^{\infty} \frac{6}{x^{\frac{5}{2}}} dx$
4. For each of the following improper integrals, find the value of the integral or **explain** briefly why it does not have a value.
- a) $\int_{27}^{\infty} \sqrt[3]{x} dx$ b) $\int_{27}^{\infty} \frac{1}{\sqrt[3]{x^4}} dx$
5. Explain briefly why $\int_0^{100} 3x^{-\frac{1}{2}} dx$ is an improper integral.
6. For each of the following improper integrals, find the value of the integral or explain briefly why it does not have a value.
- a) $\int_0^{100} 3x^{-\frac{1}{2}} dx$ b) $\int_0^{100} 3x^{-\frac{3}{2}} dx$ c) $\int_{100}^{\infty} 3x^{-\frac{1}{2}} dx$

f) Volumes of revolution

Example 20

Find the volume obtained when the shaded region is rotated through 360° about the x -axis.



$$V = \int_1^4 \pi y^2 dx = \pi \int_1^4 \frac{1}{x^2} dx$$

$$= \pi \left[-\frac{1}{x} \right]_1^4$$

$$= \pi \left[\left(-\frac{1}{4} \right) - (-1) \right]$$

$$= \frac{3}{4} \pi$$

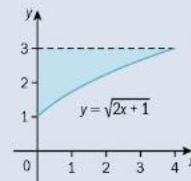
$y^2 = \left(\frac{1}{x} \right)^2$ and $a = 1$, $b = 4$

π is a constant so can be taken outside the integral.

You can leave your answer in terms of π .

Example 21

Find the volume obtained when the shaded region is rotated through 360° about the x -axis.



Volume required = volume of cylinder, with $r = 3$ and $h = 4$, minus volume of revolution about x axis.

Work out a strategy for finding the required volume.

$$y = \sqrt{2x+1}$$

$$V = \pi \int_0^4 y^2 dx = \pi \int_0^4 (2x+1) dx$$

Don't forget to square y and put values of x for the bounds.

$$= \pi [x^2 + x]_0^4$$

$$= \pi [(4^2 + 4) - (0^2 + 0)]$$

Show the substitutions clearly.

$$= 20\pi$$

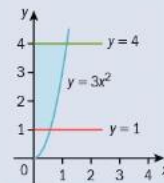
This gives us the volume when the area **below** the curve is rotated. The required volume is the volume of the cylinder found when rotating the line $y = 3$ about the x -axis with the volume of revolution removed. The volume of the cylinder is $\pi r^2 h$ where $r = 3$ and $h = 4$.

$$\text{Required volume} = \pi \times 3^2 \times 4 - 20\pi = 16\pi$$

Leave answer in terms of π unless told otherwise.

Example 22

The diagram shows the line $y = 1$, the line $y = 4$ and part of the curve $y = 3x^2$.
The shaded region is rotated through 360° about the y -axis.
Find the exact value of the volume of revolution obtained.



$$y = 3x^2, x^2 = \frac{y}{3}$$

Rearrange to $x^2 = \dots$

$$V = \int_a^b \pi x^2 dy = \pi \int_1^4 \frac{y}{3} dy$$

Use the correct formula.

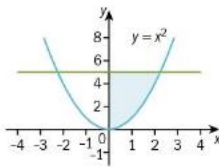
$$= \pi \left[\frac{y^2}{6} \right]_1^4$$

$$= \pi \left[\left(\frac{16}{6} \right) - \left(\frac{1}{6} \right) \right]$$

$$= \frac{15}{6} \pi = \frac{5\pi}{2}$$

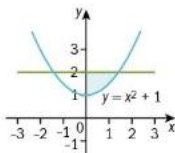
The question asked for the exact answer so leave the answer in terms of π .

3.



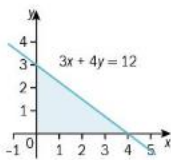
Find the volume obtained when the shaded region is rotated through 360° about the y -axis.

4.



Find the volume obtained when the shaded region is rotated through 360° about the y -axis.

5.

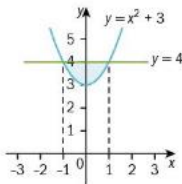


Find the volume obtained when the shaded region is rotated through 360° about the x -axis.

Fin
cor

6. Find the volume obtained when region bounded by the curve $y = x(3 - x)$ and the x -axis is rotated through 360° about the x -axis.

7.



Find the volume obtained when the shaded region is rotated through 360° about the x -axis.

Summary:

Integrating x^n and ax^n

- $\int x^n dx = \frac{x^{n+1}}{n+1} + c$, provided $n \neq -1$.
- $\int ax^n dx = \frac{ax^{n+1}}{n+1} + c$, provided $n \neq -1$.

Integrating $\int (ax + b)^n dx$

- $\int (ax + b)^n dx = \frac{1}{a} \frac{(ax + b)^{n+1}}{(n+1)} + c$, provided $n \neq -1$.

The definite integral

- $\int_a^b f'(x) dx = [f(x)]_a^b = f(b) - f(a)$

Improper integrals

- When we have a definite integral with an upper limit of ∞ or a lower limit of $-\infty$ then we have an **improper** integral.
- When we have an integral where the function to be integrated is not defined at a point in the interval of integration then we have an **improper** integral.
- We define the improper integral $\int_a^\infty f(x) dx$ as $\lim_{b \rightarrow \infty} \int_a^b f(x) dx$, provided the limit exists.
- We define the improper integral $\int_{-\infty}^b f(x) dx$ as $\lim_{a \rightarrow -\infty} \int_a^b f(x) dx$, provided the limit exists.
- When $f(x)$ is defined for $0 \leq x < b$, but $f(x)$ is not defined when $x = 0$, then the **improper integral** $\int_0^b f(x) dx$ is defined as $\lim_{a \rightarrow 0^+} \int_a^b f(x) dx$, provided the limit exists.

Area under a curve

- The area under the curve $y = f(x)$ bounded by the x -axis, the line $x = a$ and the line $x = b$ is given by:

$$\text{Area} = \int_a^b y \, dx = \int_a^b f(x) \, dx$$

If the result is positive the region lies above the x -axis.

If the result is negative the region lies below the x -axis.

- The area under the curve $x = f(y)$ bounded by the y -axis, the line $y = a$ and the line $y = b$ is given by:

$$\text{Area} = \int_a^b x \, dy$$

If the result is positive the region is to the right of the y -axis.

If the result is negative the region is to the left of the y -axis.

Volume of revolution

- If an area bounded by a curve is rotated through 360° about the x -axis we can write

the volume of the solid formed is $V = \int_a^b \pi y^2 \, dx$, where a and b are the boundary values for x .

- If an area bounded by a curve is rotated through 360° about the y -axis we can write

the volume of the solid formed is: $V = \int_a^b \pi x^2 \, dy$, where a and b are the boundary values for y .